**Fall 2025**

**CSE400C**

Capstone Project

# Project Title: Exploring Fusion Strategies for Multimodal Emotion Recognition Using BERT, Bidirectional LSTMs, and Audio Feature Extractors


## Submitted By:

| Student Name | Student ID |
|---|---|
| Rifa Azad | 2022-1-60-265 |
| Tahura Akter Tripty | 2022-1-60-051 |
| Fahimun Islam Lamia | 2021-3-60-181 |

## Project Supervisor:

**Dr. Shamim H Ripon**

Professor

Department of Computer Science and Engineering


**Date of Submission: 11 December 2025**

# Table of Contents

Abstract ..... 4
1. Introduction ..... 4
1.1 Problem Statement ..... 5
1.2 Research Questions ..... 5
1.3 Objectives ..... 6
1.4 Focus of the Study ..... 6
1.5 Contributions ..... 7
1.6 Outline ..... 7
2. Background ..... 7
2.1 Multimodal Fusion Strategies ..... 7
2.2 Related Works: ..... 9
2.2 Table 1: Summary of related works ..... 13
3. Materials and Method ..... 16
3.1 Techniques ..... 16
3.2 Skills ..... 17
3.2.2 Materials and Devices ..... 17
3.3 Dataset Description ..... 18
3.3.1 Dataset Overview ..... 18
3.3.2 Emotion Categories and Data Processing ..... 18
3.3.3 Data Selection and Sample Statistics ..... 18
3.3.4 Speaker Distribution and Cross-Validation Strategy ..... 19
3.3 System Design ..... 19
4. Design and Implementation ..... 20
4.1 Audio-Only Emotion Recognition System ..... 20
4.1.1 Audio Preprocessing and Feature Extraction ..... 20
4.1.2 Bidirectional LSTM with Self-Attention Architecture ..... 21
4.1.3 Training Strategy and Optimization ..... 22
4.1.4 Convolutional Neural Network Baseline ..... 22
4.2 Text-Only Emotion Recognition System ..... 22
4.2.1 Text Preprocessing ..... 22
4.2.2 BERT-Based Text Encoder ..... 23
4.2.3 Training Configuration ..... 23

4.3 Multimodal Fusion Architectures ..... 24
4.3.1 Early Fusion: Feature-Level Concatenation ..... 24
4.3.2 Hybrid Fusion: Combining Early and Late Strategies ..... 24
4.3.3 Late Fusion: Adaptive Gating ..... 25
4.3.4 Cross-Modal Attention Fusion ..... 25
4.3.5 Late Fusion: Performance-Based Weighting ..... 25
4.3.6 Late Fusion: Learnable Weight Optimization ..... 26
5. Result Analysis ..... 26
5.1 Audio-Only Models ..... 26
5.1.1 Bidirectional LSTM Performance ..... 26
5.1.2 CNN Baseline Performance ..... 28
5.2 Text-Only Models ..... 28
5.2.1 BERT Fine-tuning Performance ..... 28
5.3 Multimodal Fusion Results ..... 29
5.3.1 Early Fusion: Feature-Level Concatenation ..... 29
5.3.2 Hybrid Fusion: Dual-Pathway Architecture ..... 31
5.3.3 Late Fusion: Adaptive Gating ..... 32
5.3.4 Cross-Modal Attention Fusion ..... 33
5.3.5 Performance-Based Weighted Fusion ..... 34
5.3.6 Learnable Wight Fusion ..... 36
5.4 Overall Comparative Analysis ..... 37
5.4.1 Performance Summary ..... 37
5.4.2 Key Findings ..... 37
5.4.3 Practical Implications ..... 38
6. Conclusion ..... 38
6.1 Summary ..... 39
6.1.1 Key Insights and Contributions ..... 39
6.1.2 Limitations ..... 40
6.1.3 Social and Environmental Impact ..... 40
6.2 Future Work ..... 41
Bibliography ..... 41
Appendix ..... 43

# Abstract

Speech Emotion Recognition (SER) plays a crucial role in human-computer interaction, yet traditional unimodal approaches face limitations in noisy environments and speaker variability. This study develops a multimodal SER system integrating acoustic and linguistic modalities using the IEMOCAP dataset. We systematically compared MFCC and log-mel spectrogram features with CNN and BiLSTM architecture for audio processing, alongside BERT-based models for text analysis. Building on optimized unimodal baselines (Audio BiLSTM: 59.23% UAR; Text BERT: 63.00% UAR), we implemented and evaluated six fusion strategies: early feature concatenation, late fusion with adaptive gating, hybrid fusion, cross-modal attention, performance-based weighting, and learnable gradient-based weighting. Results demonstrate that Late Fusion with Adaptive Gating achieves the highest performance (70.60% UAR, 68.85% WA, 69.06% F1), representing a 7.60% improvement over the best unimodal baseline. Cross-modal attention (69.49% UAR) and early fusion (68.29% UAR) also showed substantial gains. Modified Leave-One-Speaker-Out cross-validation with session separation constraints ensured speaker-independent generalization. The best-performing model was deployed as a Streamlit web application, demonstrating practical real-time emotion prediction capabilities.


# 1. Introduction

Speech is a fundamental communication method that conveys feelings, thoughts, and emotions through words. Among various modalities for expressing emotions, including facial expressions, text, and physiological signals, speech (audio) remains the most natural and commonly used medium. This intrinsic relationship between speech and emotion has led to the development of Speech Emotion Recognition (SER) systems, where artificial intelligence technology is utilized to identify human emotions expressed in spoken statements.
SER has diverse practical applications including healthcare for mental health monitoring, call centers for customer satisfaction analysis, online streaming media for content personalization, and smart assistants for home automation and remote area monitoring. These applications significantly impact human-computer interaction, making SER an important research area. However, human perception of emotions can be biased due to factors like mood, accent, or language. Similarly, SER systems face real-world challenges such as background noise and low-quality audio recordings.

Traditional SER systems initially relied on machine learning models with manually selected audio features such as pitch, tone, and spectrograms. However, these approaches faced limitations due to uncertainty in selecting the most relevant features for accurately identifying different emotions. The field experienced a transformation with the advancement of deep learning. Researchers have shown preference for deep learning methods because models such as CNNs, LSTMs, BiLSTMs, and transformers demonstrate higher performance in SER by automatically learning complex acoustic and temporal patterns directly from raw or spectrogram-based features. Further progress came through hybrid models and transfer learning approaches, particularly in real-world conditions.

Despite these advances in audio-based SER, critical insight emerged: emotions are encoded not only in *how* something is said (audio prosody) but also in *what* is said (textual content). This realization has led to the rise of multimodal SER, which integrates both speech audio and spoken text to enhance emotional understanding.

Aligned with this direction, this capstone project aims to build a comprehensive multimodal SER system using the IEMOCAP dataset containing both audio recordings and their corresponding transcripts. The project systematically investigates optimal configurations for both modalities: comparing MFCC and log-mel spectrogram features combined with CNN and BiLSTM architectures for audio processing and implementing BERT-based embeddings for text analysis. Building upon these optimized unimodal models, we implement and rigorously compare six fusion strategies (early feature-level fusion, late fusion with adaptive gating, hybrid fusion combining both early and late approaches, cross-modal attention fusion, performance-based weighting, and learnable gradient-based weighting) to determine the most effective approach for integrating acoustic and linguistic information. Finally, the best-performing model will be deployed as a user-friendly web application for real-time emotion prediction.

## 1.1 Problem Statement

Speech Emotion Recognition (SER) plays a crucial role in human-computer interaction by accurately identifying emotional states from spoken language. However, most existing SER systems rely on audio-only input, which often struggles under noisy conditions or with speaker variation. Although deep learning models have improved performance in emotion recognition, relying exclusively on acoustic features limits emotional understanding.
Recent research demonstrates that combining audio with text can significantly improve emotion recognition. Despite promising results, multimodal approaches remain underutilized in practical applications, and critical questions about optimal fusion strategies persist. Most studies evaluate single fusion approaches without systematic comparison, employ inconsistent evaluation protocols, or neglect rigorous speaker-independent validation essential for real-world deployment.

To address these challenges, this capstone project proposes the development of a multimodal SER system using the IEMOCAP dataset. The project systematically investigates traditional fusion paradigms (early, late, hybrid, cross-modal attention) and also explores adaptive weighting mechanisms (performance-based and learnable gradient-based weighting) to optimize modality integration, under controlled experimental conditions to identify the optimal strategy for integrating audio and text modalities. Additionally, we implement a modified Leave-One-Speaker-Out (LOSO) cross-validation with session separation constraints to ensure realistic performance estimates. The best-performing model will be deployed as an accessible web application, demonstrating practical applicability for real-world emotion recognition tasks.

## 1.2 Research Questions

The research questions guiding this investigation are:

1: Which feature extraction technique and neural architecture combination yields the best results in audio-based emotion classification?

2: Which fusion strategy achieves superior performance when integrating audio and text modalities for emotion recognition?

3: How does the performance of the best multimodal fusion approach compare to unimodal baselines (audio-only BiLSTM and text-only BERT) in terms of emotion classification accuracy?

4: How can the best-performing model be deployed as a user-friendly application for real-time emotion prediction?

## 1.3 Objectives

The primary objective of this capstone project is to develop a multimodal speech emotion recognition system using the IEMOCAP dataset by integrating audio and text features to enhance emotion classification accuracy.
The specific objectives are:

- To investigate and evaluate various feature extraction techniques for audio-based emotion recognition and determine the most effective approach for capturing emotional characteristics from speech signals.
- To develop and compare deep learning architectures(specifically CNNs with channel attention and BiLSTMs with self-attention) for audio feature processing, alongside BERT-based models for text-based emotion classification.
- To implement and systematically compare six fusion strategies (early feature-level fusion, late fusion with adaptive gating mechanisms, hybrid fusion combining early and late approaches, cross-modal attention fusion, performance-based weighting, and learnable gradient-based weighting) for combining audio and text modalities in a multimodal SER system.
- To deploy the best-performing model as a user-friendly web application for real-time emotion prediction, demonstrating the practical applicability of the developed system.

## 1.4 Focus of the Study

This study investigates multimodal emotion recognition by comparing several fusion strategies that combine acoustic and linguistic information from speech. The research emphasizes speaker-independent evaluation using a strict LOSO cross-validation setup with session separation, ensuring realistic generalization to unseen speakers. It systematically benchmarks a wide range of fusion approaches, from early feature-level fusion to hybrid models and multiple late fusion strategies, under identical conditions to identify which integration methods are most effective. The study also prioritizes practical interpretability by examining modality contributions and fusion behaviors, focusing on when and why the model relies on audio or text. By balancing architectural complexity with limited training data, this work aims to offer both theoretical insight and practical guidance for developing a deployable multimodal emotion recognition system.

## 1.5 Contributions

This research provides a comprehensive comparison of six multimodal fusion strategies for emotion recognition, establishing strong baselines for audio-only, text-only, and fused systems on the IEMOCAP dataset. It offers new insights into modality complementarity by showing how optimal audio–text weighting varies across emotion categories and how fusion improves speaker-independent robustness. Methodologically, the study contributes a rigorous LOSO protocol with session separation and demonstrates effective handling of class imbalance through weighted training and UAR evaluation. It highlights important architectural findings, showing that simple fusion mechanisms such as adaptive gating can outperform more complex models under data constraints. The work also delivers a reproducible experimental pipeline and interpretable fusion analyses, providing practical tools and evidence that lightweight fusion models can achieve deployment-ready performance.

## 1.6 Outline

The paper highlights the Introduction in Chapter 1, the Related Works in Chapter 2, the Materials and Methods in Chapter 3, the Implementation in Chapter 4, the Results and Analysis in Chapter 5, the Conclusion in Chapter 6, and then followed by the Bibliography and the Appendix.

# 2. Background

Speech Emotion Recognition (SER) has gained increasing attention as a means of improving human-computer interaction by enabling systems to detect emotions from speech. Traditional SER models primarily relied on handcrafted audio features such as pitch, tone, and Mel-Frequency Cepstral Coefficients (MFCCs). However, these unimodal approaches often struggled with noisy environments, speaker variability, and ambiguous expressions.

To address these limitations, recent studies have adopted deep learning techniques such as CNNs, LSTMs, and Transformers, which learn complex acoustic and temporal patterns directly from data. Despite these improvements, focusing solely on audio leaves important emotional cues unexplored, as the meaning of speech also depends on textual content. Emotions are encoded not only in *how* something is said (prosody, tone, rhythm) but also in *what* is said (word choice, semantic content, context).

To overcome this limitation, multimodal SER systems integrate both acoustic and textual information. Our work designs such a system based on the IEMOCAP dataset, which provides synchronized audio recordings and speech transcripts. We compare MFCC and log-mel spectrogram features using CNN and BiLSTM architectures for audio, and employ BERT-based embeddings to capture semantic and pragmatic emotional cues from text. Most importantly, we systematically evaluate three fusion mechanisms to determine the optimal strategy for integrating these complementary modalities.

## 2.1 Multimodal Fusion Strategies

Multimodal fusion can be categorized into four main approaches, distinguished by when and how information from different modalities is combined:

**Early Fusion (Feature-Level Fusion)** combines features from different modalities at the input level before feeding into a unified model. This approach concatenates processed audio and text features into a single feature vector, allowing the model to learn joint representations that capture correlations between acoustic prosody and linguistic semantics. The advantage is maximal information integration through deep cross-modal interactions. However, early fusion faces challenges including high dimensionality, potential modality dominance where one modality overwhelms the other during training, and difficulty handling missing modalities during deployment.

**Late Fusion (Decision-Level Fusion)** trains separate models for each modality independently and combines their predictions at the decision level. This approach maintains modality-specific optimization, enabling specialized architectures tailored to each information stream (CNNs for audio spectrograms, Transformers for text). Predictions can be combined through simple averaging, weighted voting, or learned meta-classifiers. The advantages include modularity, robustness to missing modalities, and independent model development. However, late fusion prevents low-level cross-modal feature interactions, potentially missing complementary patterns discoverable through joint learning.

**Performance-Based Weighting** assigns fusion weights based on individual modality performance metrics (e.g., validation accuracy or F1-scores). Unlike fixed averaging in traditional late fusion, this approach explicitly accounts for the reliability of each modality by giving higher weights to better-performing models. For instance, if text BERT achieves 63% UAR while audio BiLSTM achieves 59% UAR, performance-based weighting would assign proportionally higher influence to text predictions. This strategy provides a principled alternative to uniform averaging while maintaining the modularity advantages of late fusion. However, static performance-based weights may not adapt to instance-specific variations in modality informativeness.

**Learnable Gradient-Based Weighting** employs trainable parameters to learn optimal fusion weights through backpropagation during joint training. Unlike fixed performance-based weights, this approach allows the model to automatically discover optimal modality combinations through gradient descent on the fusion objective. Learnable weights can theoretically adapt to training data characteristics and capture complex modality interactions beyond simple validation performance metrics. However, this approach requires careful regularization to prevent overfitting and may be sensitive to training dynamics and initialization.

**Hybrid Fusion** combines both early and late fusion approaches to leverage advantages of both paradigms. This strategy typically processes modalities through separate pathways initially (maintaining modality-specific optimization like late fusion), then combines intermediate representations at a feature level (enabling cross-modal interactions like early fusion), before making final predictions. Hybrid fusion can capture both low-level feature correlations and high-level decision-making patterns, offering a middle ground between the extremes of pure early or late fusion. However, it introduces additional architectural complexity and requires careful design of where and how to merge modality-specific representations.

**Cross-Modal Attention Fusion** uses attention mechanisms to enable dynamic, context-dependent modality weighting and interaction at intermediate representation levels. Attention layers allow the model to adaptively emphasize relevant modalities for specific instances, prioritizing acoustic features during prosodically salient segments while weighting linguistic content during semantically-rich passages. Cross-attention mechanisms further enable bidirectional interaction where audio and text representations can attend to each other, facilitating fine-grained correspondence discovery. This flexibility offers potential for superior performance but introduces architectural complexity and increased computational requirements. Recent research consistently reports that multimodal approaches outperform unimodal baselines, with improvements ranging from 3-15% depending on datasets and fusion strategies.

This project systematically implements six fusion paradigms under controlled conditions using identical unimodal models (audio BiLSTM with log-mel spectrograms achieving 59.23% UAR; text BERT achieving 63.00% UAR). Our empirical evaluation reveals that late fusion with adaptive gating achieves the highest performance (70.60% UAR, +7.60% improvement), followed by cross-modal attention (69.49% UAR) and early fusion (68.29% UAR), providing empirical evidence for fusion strategy selection in practical SER system design.

## 2.2 Related Works:

Emotion Recognition does not solely rely on psychological and linguistic features but also considers pitch, tone, gender, and ethnicity. Numerous studies have explored these factors by applying deep learning methods with various models, fusion techniques, and transfer learning frameworks. This section will provide a brief review of the literature relevant to our study.

To show practical uses, this paper [2] introduces a novel intelligent assistant designed to identify human emotions through voice recognition and control peripherals for alert actions using a buzzer and LED alerts, with the capability of detecting seven emotions. The designed system was built on a multilayer neural network, achieving up to 85.12% accuracy while reducing error rate and processing time compared to existing technologies Gaussian Mixture Modeling (GMM) and Support Vector Machine (SVM) Model.

This paper [8] explored whether the fundamental frequency of speech alone is sufficient for emotion recognition. They used the AGH Emotional Speech Corpus, where they examined 280 recordings of 7 emotions from 12 speakers, because of its size, diversity of gender or profession, and availability. The lowest frequency was extracted via Fast Fourier Transformation (FFT) with 20 ms frames (50% overlap) for computational efficiency and noise robustness. Statistical features were derived from low frequency to assess the lowest frequency contours to capture pitch patterns. Five WEKA classifiers were tested for low-frequency discriminatory power across emotion pairs. It was chosen as a computationally cheap, well-established acoustic correlation of emotion, and provides a reliable benchmark for assessing the performance of future multi-feature systems. Achieving up to 89.74% accuracy for two emotions, the findings establish a strong baseline for future emotion recognition systems.

Paper [9]implemented a CNN-based model for real-time emotion recognition on the RAVDESS

dataset. The system categorized emotions into four (happy, sad, neutral, and disgust) and achieved an accuracy of 85%. The study focuses on the importance of Discrete Wavelet Transform (DWT) for feature extraction and addresses deployment challenges like background noise and model efficiency in smart assistants and embedded systems. Future directions are harmonious with advancement in multimodal AI.

Paper [10] uses CNN-LSTM and Attention-Enhanced CNN-BiLSTM models for speech emotion recognition on the RAVDESS dataset. The authors transform speech signals into Mel spectrograms and pass them into time-distributed CNN layers to extract spatial features. Then use LSTM layers to model temporal dependencies. The second model upgrades this with bidirectional LSTM and an attention mechanism to focus on emotionally relevant frames. Data augmentation using uniform noise and additive white Gaussian noise (AWGN) was applied to improve generalization. The models achieved 96.5% accuracy for CNN-LSTM and 98.1% for the attention-enhanced version.

In paper,[11] the BanglaSER dataset participants expressed five emotions, which are angry, happy, neutral, sad, and surprise. Human evaluators achieved an accuracy of 80.5%, with the most frequent misclassification observed between surprise and happiness, which accounted for 13.5% of the errors. Experimental results showed that MFCC features achieved 60.86% accuracy, epoch-based features 54.52% while their fusion resulted in an accuracy of 65.93%, reflecting a 5.07% improvement. The remaining 14.52% were below human performance, highlighting the necessity for more advanced feature engineering, data augmentation, and temporal modeling to bridge the gap.

This paper[12] develops an emotion recognition system using the RAVDESS dataset, comparing audio features and deep learning modes. Log-Mel spectrograms are prioritized over MFCCs because they better mimic human auditory processing and retrain critical pitch or energy patterns. A 4-layer 2D CNN was selected for its spatial feature extraction from the spectrogram, while global average pooling handled variable audio lengths to reduce overfitting. Gender specific classification was adopted due to biological acoustic differences between voices. This approach achieved 68% accuracy for RAVDESS and 62% for TESS. Demonstrating that feature engineering outperforms model complexity in speech emotion recognition.

It is important to know the divergence between traditional and deep learning methods for SER before leaping towards a specific path, which has been explored in this paper. The paper[13] traces the evolution of SER algorithms starting from conventional ML methods that require extensive feature engineering and deep domain knowledge to deep learning, emphasizing LSTM-CNN hybrids.

Paper[14] conducted an extensive survey of traditional and deep learning approaches in SER and shows how deep learning approaches outperform classical ML models like SVM, k-NN. Deep learning models like CNN, RNN, LSTM, and GRU were highlighted for their ability to learn complex emotional features automatically. It emphasizes the use of MFCC, Spectrograms, discusses public datasets like RAVDESS, TESS, and EmoDB, and suggests hybrid deep learning models for the future.

To enhance the real-time SER system, this paper[5] used CNN + BiLSTM with data augmentation techniques on RAVDESS, TESS, and EmoDB datasets. Using features like MFCCs, Mel spectrograms, ZCR, RMS, and chroma with data augmentation( noise injection, pitch shifting), the model achieved top accuracies of 100% (TESS), 99.5% (EmoDB), and 90.12% (RAVDESS). This proved that combining spatial and temporal audio features with augmented training superior performance in real-time scenarios and noise-resilient SER systems.

Paper[4] uses a two-stream CNN-BiGRU model on the KEMDy20 dataset and uses both MFCC and Mel-spectrogram features as input. The model allows one stream to process MFCC and the other to process spectrograms and merge them before classification. Achieving 89.71% accuracy, this paper signifies that using multi-feature fusion and recurrent layers improves emotion recognition performance.

Paper[6], [15] proposes a DNN-HMM-based SER system to address the adaptability of speaker variation by combining MFCC with epoch-based features, achieving 65.93% accuracy, an increase rate of 7.13% surpassing existing cutting-edge approaches, suggesting future work with an LSTM network. Their system used a multimodal IEMOCAP dataset to emphasize the importance of deep learning methods in SER.

Hybrid models have shown a significant improvement on both language-dependent and language-independent datasets (RAVDESS, Emo-DB) in this paper[16], [17]. The proposed method of LSTM with the use of Multi-head Attention from Transformers has shown an accuracy of 71.43% on weighted accuracy and 72.50% on unweighted accuracy on the language independent dataset that was pre-processed into spectrograms from which MFCC were extracted.

Paper[6] introduced a hybrid DL model integrating two parallel CNNs and a Transformer Encoder on the RAVDESS dataset. The authors extract MFCC features from speech signals and treat them as grayscale images, which are input into CNNs for spatial feature learning. A Transformer Encoder is used to capture global temporal dependencies over speech progression. Using Additive White Gaussian Noise (AWGN), data augmentation was performed to expand the dataset. The model achieved 80.46% accuracy, showing improved recognition over the baseline DCNN model (71.61%) and proposing future integration of text-audio features to enhance classification further.

In this paper[16] a lightweight speech emotion recognition system uses a Temporal Convolution Network to process raw speech signals directly. Therefore, it uses a fast network (TCN) on raw sound, because TCN handles long audio patterns better than other networks, enabling real-time use. This means it learns emotions directly in three steps. Then, the final layers combine sound details and timing. Because they focused on speed, it scored 78.34% (IEMOCAP) and 91.61% (EMO-DV) accuracy. Therefore, this helps create practical emotion tools for robots and healthcare.

By highlighting the hindrance in SER development, this study[3] explored TL with embedding models of d-vector, enhanced x and r-vector, with image classification models. As our interest lies with SER, we have focused on the speaker embedding models integrating Multi-Head Attention pooling and Angular margin SoftMax for improved performance, reflected in the

results with an accuracy of 71.08%in unweighted accuracy and 70.97% in weighted accuracy in the enhanced r vector model. This work has revealed TL to be more effective than data augmentation, significantly outperforming models trained more scratch.

This paper[18] shows that using sound picture tricks with TL on smart networks helps computers learn feelings from speech better when there is little data. Inspired by simple emotional trees, they made a step-by-step system using sound and words. So, mixing sound with word understanding gave 74.5% accuracy on IEMOCAP. They say IEMOCAP is now used more than EMO-DB because it's bigger and real-life.

Paper[19] focuses on the theoretical study of multimodal emotion recognition without implementing any actual models. Used MFCC for audio features extraction and CNN for facial features detection. Though no model was evaluated, It highlights the role of accurate feature extraction and the necessity of combining both audio and facial modalities for improving SER systems.

Paper[20] implemented a 3D CNN-based model for speech emotion recognition using EMO-DB and eNTERFACE05 datasets. Reconstructed Phase Space (RPS) analysis is used to transform raw voice signals into 3D tensors, effectively capturing nonlinear emotional features. These tensors are used as input to a 3D CNN model consisting of multiple convolutional, pooling, and dense layers with ReLU and SoftMax activation. The model achieved a high accuracy of 92.02% for EMO-DB and 83.54% for eNTERFACE05 in speaker-independent setups. The study avoids handcrafted features but manifests limitations in computational cost, sensitivity to hyperparameters ($\tau$, d), and dataset validity.

Paper[21] built a system recognizing feelings from both speech and face. Therefore, it used two smart pre-trained networks, which are Audioset and AffectNet. Because they save time by using advanced networks. To understand changing feelings in videos, they used a special network to track the time. Finally, they mixed the sound and result together. This approach worked well, scoring 80.08% accuracy on the RAVDESS dataset for 8 emotions.

In paper the researchers created MELD, a dataset from Friends dialogues, which addresses the lack of group conversation data for emotion recognition. Also, they used features and text analysis to capture vocal and word cues. Moreover, they chose the DialogueRNN model. As a result, it improved understanding of short emotional lines where words alone aren't clear. This approach tackled key challenges like context, changing emotions, and speaker interactions.

This[22] research enhances sentiment detection by fusing text and audio data from YouTube reviews. This study achieved superior sentiment classification by integrating CNN and LSTM models to analyze text and audio data. The Bi-LSTM architecture outperformed others, reaching a notable accuracy of 94.97% in distinguishing positive and negative sentiments. This result shows the model's effectiveness in capturing complex emotional cues, implementing an efficient tool for accurate audience reaction prediction, and enhancing decision-making for content strategies.

This research[23] enhances sentiment detection by fusing text and audio data from YouTube reviews. This study achieved superior sentiment classification by integrating CNN and LSTM models to analyze text and audio data. The Bi-LSTM architecture outperformed others, reaching a notable accuracy of 94.97% in distinguishing positive and negative sentiments. This result shows the model's effectiveness in capturing complex emotional cues, implementing an efficient tool for accurate audience reaction prediction and enhancing decision-making for content strategies.

This study[7] introduced novel multimodal models for speech emotion recognition. The audio only model (ARE) was the weakest performer at 54.6% accuracy. The textual model (TRE) improved results to 63.5%. However, the best model was the Multimodal Dual Recurrent Encoder (MDRE), which used both audio and textual features to achieve an advanced accuracy of 71.8%. This shows that multimodal fusion is essential for accurately classifying emotions into angry, happy, sad, and neutral categories.

M2FNet[24] introduces a multi-modal fusion model that applies text, audio, and visual inputs through parallel attention layers. It achieves best performance, with weighted F1 scores of 66.71% on MELD and 69.86% on IEMOCAP, outperforming existing text-based and multimodal methods in emotion recognition in conversation.

This survey[25] reveals that multimodal emotion recognition significantly outperforms unimodal approaches, with accuracy improvements ranging from 2.35% to 19.73%. Effective models utilize deep learning (DBN, CNN, LSTM, Transformer) for feature extraction and apply fusion methods at the feature level, apply fusion at the decision stage, or combine audio, visual, textual, and physiological signals. These integrated systems achieve higher robustness in classifying both discrete emotions and dimensional states like valence and arousal.

## 2.2 Table 1: Summary of related works

| Ref | Title | Dataset | Model | Feature | Result | Emotion Classified |
|---|---|---|---|---|---|---|
| [1] | Speech Emotion Recognition Systems: A Comprehensive Review of Different Methodologies | Multiple public datasets (RAVDESS EmoDB IEMOCAP Regional language dataset) | 1.CNN Based<br>2. RNN / LSTM<br>3.Hybrid Model: CNN-LSTM | 1.MFCC<br>2.TEO & TMFCC<br>3.DWT<br>4.Hybrid Method: Fusion of MFCC with MS or TEO | Traditional ML: 97%<br>Deep Learning:<br>• 95.89% with 2D CNN-LSTM •92.7% with TMFCC + Deep Neural Network (DNN) | Emotions: Basic (anger, happiness, sadness) and complex (contempt, pride) emotions. |
| [2] | Multilayer Neural Network Based Speech Emotion Recognition for Smart Assistance | RAVDESS<br>TIMIT<br>Emo-DB | Multilayer Neural Network GMM, SVM, *MLP based on ANN | MFCC, MLP classification | 81.02% (RAVDESS)<br>84.23% (TIMIT)<br>85.12% (Emo-DB) | Worry, Love, Neutral, Hate, Surprise, Sadness, Happiness<br>(7 emotions) |
| [3] | Speech Emotion Recognition Using Transfer Learning: Integration of Advanced Speaker | IEMOCAP, CREMA-D | Speaker Embedding Models:<br>d-vector, | Pre-trained speaker embeddings | IEMOCAP: Enhanced r-vector:<br>UA- 71.08%<br>WA- 70.97% | Anger, Happiness (happiness & excitement combined), Sadness, and Neutral<br>(4 emotions) |

| | | | | | | |
|---|---|---|---|---|---|---|
| | Embeddings and Image Recognition Models | | x-vector (enhanced), & r-vector (enhanced) | | | |
| [4] | Speech Emotion Recognition Based on Two-Stream Deep Learning Model Using Korean Audio Information | Korean Speech Emotion Database | 1.Bi-LSTM<br>2.YAMNet<br>3.Two stream model (Bi-LSTM + YAMNet) | 1D Features: MFCC, GTCC<br>2D Spectrograms: ERB Spectrograms, Log-Mel Spectrogram. | Two-Stream Model: 96%<br>Bi-LSTM: 90.38%<br>YAMNet(CNN based): 94.91% | happy, neutral, angry, sad, chagrin, disgust, fear, and surprise (8 Emotions) |
| [5] | Real-time speech emotion recognition using deep learning and data augmentation | TESS<br>EmoDB<br>RAVDESS | 1.Hybrid model: CNN-BiLSTM<br>2.MLP<br>3.CNN | 1.MFCCs<br>2.Mel Spectrogram<br>3.ZCR<br>4.RMS<br>5.Chroma | TESS: 100%<br>EmoDB: 99.5%<br>RAVDESS: 90.12% | TESS: 6 emotions (happy, sad, angry, fearful, disgust, surprise)<br>EmoDB: 7 emotions (anger, boredom, disgust, anxiety/fear, happiness, sadness, neutral)<br>RAVDESS: 8 emotions (neutral, calm, happy, sad, angry, fearful, disgust, surprise) |
| [6] | Speech Emotion Recognition Algorithm Based on Deep Learning Algorithm Fusion of Temporal and Spatial Features | RAVDESS including AWGN-augmented data | Two parallel CNNs + Transformer Encoder | MFCCs (Mel-frequency cepstral coefficients), input as grayscale images (40×282) | 80.46% | neutral, calm, happy, sad, angry, fearful, disgust, surprised (8 emotion) |
| [7] | Multimodal Speech Emotion Recognition Using Audio and Text | IEMOCAP | ARE (Audio Recurrent Encoder), TRE (Text Recurrent Encoder), MDRE (Multimodal Dual Recurrent Encoder), MDREA (MDRE with Attention), MDRE-ASR (with ASR transcripts) | MFCCs, Prosodic Features, Word Embeddings (Text), MFCCs, Prosodic Features + Word Embeddings, MFCCs, Prosodic Features + Word Embeddings + Attention, MFCCs, Prosodic Features + ASR Word Embeddings | Best: 71.8%<br>Worst: 54.6% | Angry, Happy, Sad, Neutral |
| [8] | Speech Emotion Recognition Based on Voice Fundamental Frequency | AGH Emotional Speech Corpus | K-Nearest Neighbors, Random Forest, Bagging, JRip, Random Subspace Method | Fast Fourier Transform, Statistical feature | RF & Bagging: 89.74% | Anger-Irony, Angry-Fear, Joy-Surprise (3 emotions) |
| [9] | Real Time Emotion Recognition Using Convolutional Neural Networks | RAVDESS | CNN | DWT | 85% | neutral, calm, happy, sad, angry, fearful, disgust, surprised (8 emotion) |
| [10] | Speech Emotion Recognition: Comparative Analysis of CNN-LSTM and Attention-Enhanced CNN-LSTM Models | RAVDESS | 1.CNN-LSTM<br>2.Attention-enhanced CNN-BiLSTM | Mel Spectrogram (with noise augmentation: AWGN / Uniform) | 96.5% ( CNN-LSTM)<br>98.1% (Attention-enhanced CNN-BiLSTM) | Neutral, Calm, Happy, Sad, Angry, Fear, Disgust, Surprise (8 emotion) |
| [11] | BanglaSER: A speech emotion recognition dataset | BanglaSER | Deep Learning | Amplitude envelope,RMS energy, Zero- | Best: 65.93% | Angry, Happy, Neutral, |

| | | | | | | |
|---|---|---|---|---|---|---|
| | for Bangla language | | (CNN, LSTMs, GRUs, Transformers) | crossing rate, Mel-spectrogram, Chromagram, MFCCs, Kurtosis | | Sad, Surprise<br><br>(5 emotions) |
| [12] | Emotion Recognition from Speech | RAVDESS | CNN | Log-Mel Spectrogram, MFCCs, pitch, energy, Deltas | -68% of RAVDESS<br>-62% TESS<br>-88% binary emotion classification | Angry, sad, Neutral, Disgust, Happy,Fearful, Surprised<br>(7 emotions) |
| [15] | TC-Net: A Modest & Lightweight Emotion Recognition System Using Temporal Convolution Network | IEMOCAP, EMO-DB | Temporal Convolution Network, Fully Connected Network, SoftMax classifier | TCN-based SER, Temporal- spatial synergy (TCN, FCN), CNNs | Unweighted Accuracy<br>IEMCAP: 78.34%<br>EMO-DB: 91.61%<br>Weighted Accuracy<br>IEMCAP: 80.84%<br>EMO-DB: 92.31% | Happiness, Anger, Sadness, Neutral, Fear, Boredom, Disgust<br><br>(7 emotions) |
| [14] | DNN-HMM-Based Speaker-Adaptive Emotion Recognition Using MFCC and Epoch-Based Features | IEMOCAP | DNN-HMM | MFCC, epoch-based features & MFCC+epoch-based features | (UA considered)<br>MFCC: 60.86%<br>Epoch-based: 54.52%<br>MFCC+Epochbased:65.93% | Angry, Happy, Neutral, and Sad<br>(4 emotions) |
| [16] | Hybrid LSTM-Transformer Model for Emotion Recognition From Speech Audio Files | RAVDESS<br>Emo-DB<br>language-independent Datasets (combined other two) | *Hybrid LSTM-Transformer (Transform Encoder Layer) | MFCC | 75.62% (RAVDESS)<br>85.55% (Emo-DB)<br>72.49% (proposed) | Anger, Happiness, Sadness, Fear, Disgust, Neutral, Boredom<br>(7 emotion) |
| [17] | Improved Speech Emotion Recognition using Transfer Learning and Spectrogram Augmentation | IEMOCAP | pre-trained ResNet34 | log-mel spectrograms, spectrogram augmentation, | Exp 1: UA- 61.61%<br>WA- 66.02%<br>Exp2: UA- 65.56%<br>WA- 65.62%<br>Exp3: UA- 64.14%<br>WA- 63.61% | Exp1: Angry, Happy, Neutral, Sad<br>Exp2: Angry, Excited, Neutral, Sad<br>Exp 3: Angry, Excited+Happy, Neutral, Sad |
| [19] | A 3D Tensor Representation of Speech and a 3D Convolutional Neural Network for Emotion Recognition | EMO-DB (535 samples:German), eNTERFACE 05 (1166 samples) | 3D CNN | 3D Tensors from Reconstructed Phase Space (RPS), τ=17 (EMO-DB), τ=31 (eNTERFACE05), d=3 | EMO-DB: (speaker-dependent 92.02%) (speaker-independent 92.48%);<br>eNTERFACE05: (speaker-dependent 83.54%) (speaker-independent 84.96%) | Anger, Boredom, Disgust, Fear, Happiness, Sadness, Neutral (7 emotions) |
| [25] | Multimodal Emotion Recognition On RAVDESS Dataset Using Transfer Learning | RAVDESS | CNN-14, AlexNet, SVM, Fine-Tuning, Spatial Transformer Network, bi-LSTM, Linear SVM | AlexNet, CNN-14, Linear SVM | CNN-14:<br>76.58% | Calm, Neutral, Happy, Sad, Angry, Fearful, Disgusted, Surprised<br>(8 emotions) |
| [21] | MELD: Multimodal Emotional Lines Dataset for Emotion Recognition in Conversations | MELD (Multimodal EmotionLines Dataset) | test-CNN, boLSTM, DialogueRNN, Global GRU, Party GRU, Emotion GRU | Text: 300D GloVe embeddings, 1D-CNN, 100D feature<br>Audio: openSMILE,6373D features, L2 based SVM | Emotion Weighted: 60.25%<br><br>Sentiment Weighted: 67.56% | Anger, Disgust, Fear, Joy, Sadness, Surprise, Neutral<br>(7 emotions) |
| [26] | A multimodal hierarchical | SAVEE<br>RAVDESS | ELMo v2 (lexical | Prosodic, VQ & Spectral Features, | 81.7% (SAVEE)<br>81.2% (RAVDESS) | Angry, Happy, Disgust, Sad, Fear, Surprised |

| | | | | | | |
|---|---|---|---|---|---|---|
| | approach to speech emotion recognition from audio and text | IEMOCAP (audio-text multimodal) | features), Hierarchical DNN | MFCC, Lexical Features (ELMO) | 74.5% (IEMOCAP) | (6 emotions) |
| [22] | Maximizing Sentiment Detection through Comprehensive Multimodal Data Fusion: Integrating CNN, RNN, LSTM | IMDb movie reviews & Audio transcripts from YouTube videos | CNN, RNN, LSTM, Bi-LSTM | Multimodal fusion of text features (from CNN) and sequential audio features (from LSTM) | Bi-LSTM achieved the highest accuracy of 94.97% for sentiment classification. | Positive, Negative |
| [23] | M2FNet: Multi-modal Fusion Network for Emotion Recognition in Conversation | MELD, IEMOCAP | M2FNet with multi-head attention fusion | Text, audio, visual fusion | MELD 66.71%, IEMOCAP 69.86% | Classify emotions such as anger, disgust, sadness, joy, surprise, fear, neutral (MELD); happy, sad, neutral, angry, excited, frustrated (IEMOCAP) |
| [24] | A review of multimodal emotion recognition from datasets, preprocessing, features, and fusion methods | eNTERFACE 05, RML, BAUM-1, MELD, IEMOCAP, SEMAINE, RECOLA, MAHNOB-HCI, DEAP | Feature-level, Decision-level, Hybrid fusion, DBN, CNN, LSTM, Transformer | Handcrafted (LBP, HOG, MFCC, prosodic) and Deep Learning-based features from audio, video, text, and physiological signals | Survey of methods; Multimodal fusion improves accuracy by 2.35% to 19.73% over unimodal | Anger, Disgust, Fear, Happiness, Sadness, Surprise. |

# 3. Materials and Method

## 3.1 Techniques

Our goal in this project is to explore and compare multiple fusion strategies for multimodal emotion recognition by integrating acoustic and linguistic information from emotional speech. For this purpose, the Interactive Emotional Dyadic Motion Capture (IEMOCAP) dataset was selected as it provides synchronized audio recordings and orthographic transcriptions with reliable emotion annotations from professional actors. The dataset was obtained through academic licensing from the University of Southern California and contains approximately 12,000 utterances across five recording sessions.

The experiments were performed using high-performance computing resources with NVIDIA Tesla V100 GPUs (32GB memory) to handle the computational demands of training deep learning models including bidirectional LSTMs and BERT-based transformers. The programming language Python (version 3.8) was selected as it is the standard language for deep learning and multimodal machine learning tasks, providing comprehensive libraries and frameworks for both audio signal processing and natural language processing.

Key libraries and frameworks employed include PyTorch (1.12) for building and training neural network architectures, Librosa (0.9.2) for audio feature extraction and signal processing, and Hugging Face Transformers (4.21.0) for implementing BERT-based text models. Additional essential libraries include Scikit-learn (1.1.2) for preprocessing and evaluation metrics, NumPy and Pandas for data manipulation, and Matplotlib for visualization.

The experiments were performed using a rigorous Leave-One-Speaker-Out (LOSO) cross-validation protocol with session separation constraints to ensure speaker-independent evaluation. Audio processing techniques such as Root Mean Square (RMS) normalization, Short-Time Fourier Transform (STFT), Mel-frequency scaling, and computation of Mel-Frequency Cepstral Coefficients (MFCCs) were applied to extract acoustic features. For text processing, minimal preprocessing techniques including annotation removal and whitespace normalization were used to preserve linguistic authenticity for BERT processing.

## 3.2 Skills

This project required strong proficiency in Python and deep learning frameworks such as PyTorch and Transformers, along with practical knowledge of Librosa for audio processing. Key skills included handling tensor operations, GPU-based training, model checkpointing, and implementing LSTM, BERT, and attention-based architectures. A solid understanding of digital signal processing, such as spectral analysis, mel-scaling, and feature normalization, was essential for extracting reliable audio features. The work also demanded expertise in natural language processing, fine-tuning transformer models, and preparing text for classification. Competence in machine learning evaluation, including cross-validation, class imbalance handling, and metrics like UAR and F1-score, ensured reliable assessment. Additionally, skills in multimodal fusion, data manipulation with NumPy/Pandas, and visualization tools were necessary to manage experiments, analyze results, and maintain a reproducible workflow.

### 3.2.2 Materials and Devices

The experimental setup for this research utilized high-performance computing resources to handle the computational demands of deep learning models for multimodal emotion recognition. All experiments were conducted using Python 3.8 as the primary programming language, leveraging PyTorch 1.12 as the deep learning framework due to its flexibility and efficient GPU utilization.

The hardware configuration consisted of NVIDIA Tesla V100 GPUs with 32GB memory, which enabled efficient training of complex neural architectures including bidirectional LSTMs and BERT-based models. The computational infrastructure supported parallel processing capabilities essential for handling large-scale audio feature extraction and transformer-based text encoding. Key software libraries employed in this research include:

- **Librosa 0.9.2** for audio signal processing and feature extraction
- **Transformers 4.21.0** (Hugging Face) for BERT model implementation and fine-tuning
- **Scikit-learn 1.1.2** for preprocessing, evaluation metrics, and class weight computation
- **NumPy 1.23.0** and **Pandas 1.4.3** for data manipulation and numerical operations
- **Matplotlib 3.5.2** and **Seaborn 0.11.2** for visualization of results and analysis

The development environment was configured with CUDA 11.6 for GPU acceleration, enabling efficient batch processing during model training and inference. All experiments were designed with reproducibility in mind, utilizing fixed random seeds and consistent data splits across all experimental configurations.

## 3.3 Dataset Description

This research utilized the Interactive Emotional Dyadic Motion Capture (IEMOCAP) database, a comprehensive multimodal corpus specifically designed for emotion recognition research. IEMOCAP represents one of the most widely adopted benchmarks in the affective computing community, providing rich annotations and diverse modalities for studying human emotional expressions.

### 3.3.1 Dataset Overview

The IEMOCAP dataset comprises approximately 12,000 utterances recorded from five professional actors (two male, three female) across multiple recording sessions. Each session involved scripted and improvised dyadic interactions designed to elicit authentic emotional expressions. The dataset provides synchronized recordings across four modalities: audio signals, video frames, motion capture data, and orthographic transcriptions.
Each utterance in the dataset received detailed annotations from at least three human evaluators, ensuring high inter-annotator reliability. The annotations include both categorical emotion labels and dimensional ratings for valence, activation, and dominance. This multi-annotator approach provides robust ground truth labels that account for the inherent subjectivity in emotion perception.

### 3.3.2 Emotion Categories and Data Processing

While the original IEMOCAP dataset includes ten emotion categories (angry, happy, sad, neutral, excited, frustrated, fearful, surprised, disgusted, and other), this research focused on four core emotion classes following established conventions in speech emotion recognition literature. The selected emotions and their mappings are:

- **Anger** (ang): Representing high-arousal negative emotions
- **Happiness** (hap + exc): Combining happy and excited states as both represent high-arousal positive emotions
- **Sadness** (sad): Representing low-arousal negative emotions
- **Neutral** (neu): Representing the absence of strong emotional content

This consolidation addresses several practical considerations. First, the reduced emotion set provides more balanced class distributions, as rare emotions like fear, surprise, and disgust contain insufficient samples for effective deep learning model training. Second, the merger of happiness and excitement reflects their psychological similarity; both represent positive valence with high activation and are frequently confused even by human annotators. Third, this four-class framework aligns with established emotion recognition research, facilitating meaningful comparisons with prior work while covering the primary emotional dimensions relevant for real-world applications.

### 3.3.3 Data Selection and Sample Statistics

To ensure high-quality spontaneous emotional expressions, this research exclusively utilized improvised dialogue segments, excluding scripted interactions. This decision prioritizes

naturalistic emotional content that better represents real-world communication scenarios. The improvised segments contain more authentic prosodic patterns, lexical choices, and emotional dynamics compared to scripted readings.

After applying the emotion mapping and improvisation filtering criteria, the final dataset comprised **2,943 utterances** distributed across the four emotion categories. The class distribution reflects natural emotional expression patterns observed in dyadic conversations:

- **Neutral**: 1,099 utterances (37.3%) - The most frequent category, representing baseline conversational states
- **Happiness**: 947 utterances (32.2%) - Including both happy and excited expressions, forming the largest positive emotion class
- **Sadness**: 608 utterances (20.7%) - Representing negative low-arousal states
- **Anger**: 289 utterances (9.8%) - The minority class, reflecting the relative infrequency of high-arousal negative emotions in conversation

This imbalanced distribution presents a realistic challenge for emotion recognition systems, as it mirrors natural emotional occurrence patterns where neutral and positive emotions dominate everyday interactions while negative high-arousal states like anger appear less frequently.

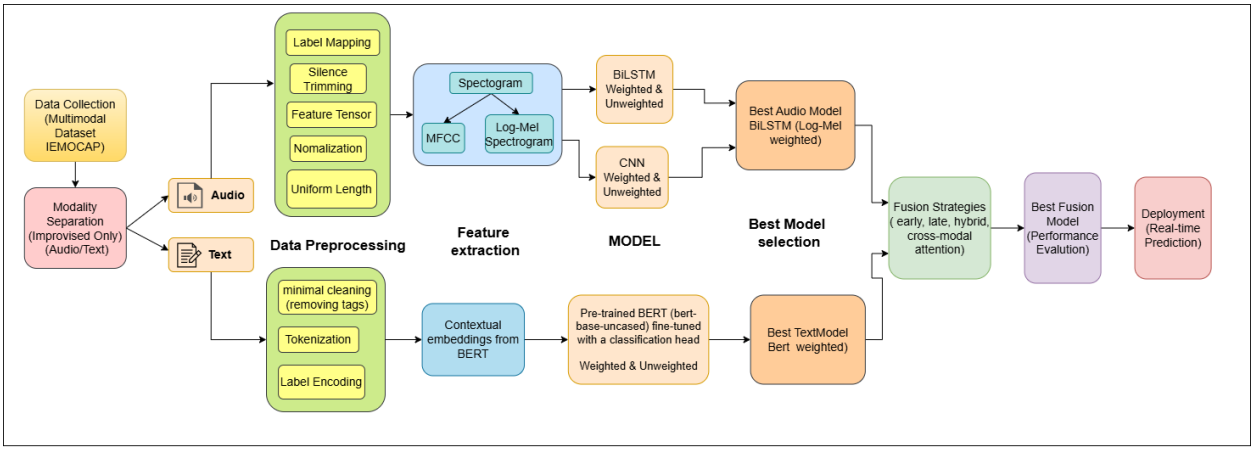
### 3.3.4 Speaker Distribution and Cross-Validation Strategy

The dataset spans recordings from ten speakers across five sessions (two speakers per session). This speaker diversity is crucial for evaluating model generalization capabilities. To ensure robust speaker-independent evaluation, this research implemented a modified Leave-One-Speaker-Out (LOSO) cross-validation protocol with strict session separation constraints.

In this evaluation framework, each fold designates one speaker as the test set while selecting a validation speaker from a different recording session than the test speaker. This constraint prevents session-specific effects, such as recording conditions, microphone characteristics, or background noise patterns, from leaking between validation and test sets.

The resulting 10-fold cross-validation structure ensures that each speaker serves as the test set exactly once, while validation speakers are always drawn from different sessions. This rigorous protocol provides unbiased estimates of model performance on completely unseen speakers, simulating real-world deployment scenarios where the system encounters individuals who do not present during training.

## 3.3 System Design




# 4. Design and Implementation

This section presents detailed architectural designs and implementation strategies for all experimental models, including unimodal baselines and multimodal fusion approaches. The methodology progresses systematically from single-modality models through increasingly sophisticated fusion strategies.

## 4.1 Audio-Only Emotion Recognition System

### 4.1.1 Audio Preprocessing and Feature Extraction

The audio processing pipeline transforms raw waveforms into standardized spectral-temporal representations suitable for deep learning models. Each audio file undergoes a multi-stage preprocessing workflow designed to normalize amplitude variations and extract emotionally salient acoustic characteristics.

**Normalization and Duration Standardization**: Raw audio signals first undergo Root Mean Square (RMS) normalization to ensure consistent amplitude scaling across utterances, compensating for recording-level variations. All utterances are then standardized to a fixed 4-second duration through zero-padding for shorter clips or center-cropping for longer segments. This duration captures complete emotional expressions while maintaining computational efficiency.

**Acoustic Feature Extraction**: The system extracts two complementary spectral representations:

1. **Log-Mel Spectrograms**: Mel-frequency scaled spectrograms with 80 mel-frequency bins, computed using 25ms Hamming windows with 10ms hop length. The logarithmic amplitude scaling compresses the dynamic range, emphasizing perceptually relevant energy patterns while reducing sensitivity to absolute intensity variations.

2. **Mel-Frequency Cepstral Coefficients (MFCCs)**: 40-dimensional MFCC features extracted with identical windowing parameters, capturing the spectral envelope

characteristics that convey emotional prosody and vocal tract configurations.

**Temporal Dynamics Enhancement**: To capture temporal evolution patterns essential for emotion recognition, the system optionally computes first-order delta coefficients (velocity) and second-order delta-delta coefficients (acceleration) for both feature types. These derivatives encode dynamic prosodic cues such as pitch contours, energy modulation, and speaking rate variations that differentiate emotional states.

**Feature Normalization**: To prevent data leakage while ensuring stable model training, the pipeline implements fold-specific global z-score normalization. Mean and standard deviation statistics are computed exclusively from training data within each cross-validation fold, then applied consistently to validation and test sets. This machine learning hygiene practice ensures genuine out-of-sample evaluation.

### 4.1.2 Bidirectional LSTM with Self-Attention Architecture

The audio emotion recognition model employs a bidirectional Long Short-Term Memory (BiLSTM) network enhanced with a self-attention mechanism, designed to capture both sequential dependencies and temporally salient emotional cues in acoustic features.

**Bidirectional LSTM Encoder**: The architecture utilizes two stacked BiLSTM layers, each with 128 hidden units per direction (256 total). Bidirectional processing captures both forward and backward temporal contexts, enabling the model to leverage future information when interpreting current time steps, crucial for understanding prosodic patterns that unfold over entire utterances. The stacked configuration allows hierarchical feature learning, where the first layer extracts low-level acoustic patterns while the second layer models higher-level emotional prosody.

**Self-Attention Mechanism**: Rather than relying solely on the final LSTM hidden state, the architecture incorporates a self-attention layer that dynamically computes important weights across all time steps. This mechanism allows the model to focus on emotionally discriminative segments—such as stressed syllables, laughter, or prosodic emphasis, while down weighting less informative regions. The attention mechanism computes alignment scores using a learned projection of the LSTM outputs, then produces a weighted context vector summarizing the most relevant temporal information.

**Classification Head**: The attended representation passes through a classification module consisting of GroupNorm (chosen over BatchNorm for stability with small batch sizes), dropout regularization (p=0.3), and a linear projection to four emotion classes. GroupNorm provides normalization benefits without depending on batch statistics, making the model robust to batch size variations common in speaker-independent validation scenarios.
**Model Capacity**: The complete architecture contains approximately 850K trainable parameters, providing sufficient capacity for capturing complex emotional prosody patterns while remaining trainable on the limited IEMOCAP dataset without severe overfitting.

### 4.1.3 Training Strategy and Optimization

**Class Imbalance Handling**: Given the natural imbalance in emotional expression frequencies, the training employs class-weighted CrossEntropyLoss computed using inverse frequency weighting. This assigns higher loss penalties to misclassifications of minority emotions (particularly anger at 9.8% of samples), preventing the model from achieving high accuracy by simply predicting majority classes.

**Optimization Configuration**: The model trains use AdamW optimizer with a learning rate of $1\times10^{-3}$ and weight decay of $1\times10^{-4}$ for regularization. Gradient clipping with a maximum norm of 1.0 prevents gradients exploding during training. Automatic mixed precision (AMP) training accelerates computation while maintaining numerical stability.
**Learning Rate Scheduling**: A ReduceLROnPlateau scheduler monitors validation loss with patience of 3 epochs, reducing the learning rate by a factor of 0.5 when performance plateaus. This adaptive scheduling enables fine-grained convergence in later training stages.

**Early Stopping**: Training implements early stopping with patience of 8 epochs based on validation loss, preserving the best model checkpoint. This prevents overfitting while ensuring sufficient training duration for convergence.

**Evaluation Metrics**: Model performance is assessed using three complementary metrics:

- **Unweighted Average Recall (UAR)**: Macro-average recall giving equal importance to all emotion classes, serving as the primary metric for imbalanced datasets
- **Weighted Accuracy (WA)**: Overall accuracy weighted by class frequencies
- **Macro F1-Score**: Harmonic mean of precision and recall across classes

### 4.1.4 Convolutional Neural Network Baseline

For comparative analysis, an alternative CNN-based architecture was implemented treating spectrograms as 2D images with frequency and time axes.

**Architecture Design**: The AudioCNN processes spectrograms through four convolutional blocks with progressively increasing filter depths (32→64→128→256) and spatial down sampling via strided convolutions. Each block contains 3×3 convolutions, batch normalization, ReLU activation, and max pooling. A channel attention module emphasizes emotionally relevant frequency bands before global average pooling and classification.

**Training Configuration**: The CNN employs OneCycleLR scheduling for faster convergence, with early stopping patience of 6 epochs and identical class weighting strategies as the LSTM model.

## 4.2 Text-Only Emotion Recognition System

### 4.2.1 Text Preprocessing

The text preprocessing pipeline applies minimal transformations to preserve linguistic authenticity crucial for BERT's contextual understanding while removing transcription artifacts.

**Annotation Removal**: Paralinguistic markers such as [LAUGH], [BREATHING], and [NOISE] are stripped as they represent transcription conventions rather than actual words processed by language models.

**Whitespace and Punctuation Normalization**: Multiple consecutive spaces are compressed to single spaces, and excessive repeated punctuation is reduced while preserving basic emotional cues conveyed through punctuation patterns.

**Authenticity Preservation**: Unlike traditional NLP preprocessing, the pipeline avoids aggressive stemming, lemmatization, or stop-word removal, maintaining natural language patterns that carry emotional content through linguistic choices, sentence structure, and writing style.

### 4.2.2 BERT-Based Text Encoder

**Model Selection**: The text emotion recognition system employs **bert-base-uncased**, a 12-layer transformer encoder pre-trained on large-scale text corpora. This 110M parameter model provides rich contextual word representations through its bidirectional self-attention mechanism, capturing semantic relationships and linguistic patterns associated with emotional expressions.

**Tokenization and Sequence Length**: Analysis of token length distributions across cleaned transcripts revealed that 95% of utterances contain fewer than 100 tokens. A maximum sequence length of 128 tokens was selected to balance computational efficiency with information preservation, ensuring complete emotional context capture without excessive padding or truncation.

**Fine-tuning Architecture**: A classification head consisting of dropout ($p=0.1$) and a linear layer maps BERT's [CLS] token representation (768 dimensions) to four emotion logits. Fine-tuning updates all BERT parameters and the classification head simultaneously, adapting pre-trained linguistic knowledge to emotional language patterns.

### 4.2.3 Training Configuration

**Optimization Setup**: The model fine-tunes for 4 epochs using AdamW optimizer with learning rate $2\times10^{-5}$, batch size 16, and linear warmup over 10% of training steps. This configuration follows established BERT fine-tuning practices, preventing catastrophic forgetting of pre-trained knowledge.

**Class Weighting**: Identical inverse frequency weighting as audio models addresses emotional imbalance, implemented through weighted CrossEntropyLoss.
**Custom Training Framework**: A WeightedTrainer extending Hugging Face's Trainer class enables class-weighted loss computation with automatic device placement. A TrainingHistoryCallback tracks training dynamics and saves the best model based on validation loss.

**Evaluation Metrics**: Text models are evaluated using UAR, weighted accuracy, and macro F1-score, maintaining consistency with audio evaluation protocols for fair multimodal comparison.

## 4.3 Multimodal Fusion Architectures

### 4.3.1 Early Fusion: Feature-Level Concatenation

**Feature Extraction**: Pre-trained audio and text models serve as fixed feature extractors. The AudioLSTM with attention outputs 256-dimensional acoustic embeddings, while BERT produces 768-dimensional textual embeddings from [CLS] tokens.

**Normalization Strategy**: Audio and text features undergo fold-specific z-score normalization computed from training data, ensuring consistent feature scales before concatenation. This prevents one modality from dominating due to different magnitude ranges.

**Fusion Architecture**: The concatenated 1024-dimensional feature vector passes through a lightweight classification head consisting of two fully connected layers (1024→512→4) with ReLU activations and dropout (p=0.3) for regularization. This design allows the fusion layer to learn complementary patterns between modalities while avoiding excessive parameters that could cause overfitting on limited data.

**Training Protocol**: The fusion classifier trains exclusively on combined features with frozen audio and text encoders, using class-weighted CrossEntropyLoss, early stopping (patience=10), and fold-specific data splits maintaining speaker independence.

### 4.3.2 Hybrid Fusion: Combining Early and Late Strategies

**Dual-Pathway Architecture**: This approach processes both raw features and probability predictions through separate pathways, then fuses their outputs.

**Early Fusion Pathway**: Processes concatenated 1024-dimensional audio-text features through a 512→256 dimensional network capturing joint low-level patterns.

**Late Fusion Pathway**: Processes log-probabilities (not raw probabilities) from individual model predictions through a parallel 8→16 dimensional network. Log-space processing avoids bounded probability constraints that can limit representational capacity.

**Fusion Strategies**: Three combination methods were evaluated:

1. **Concatenation**: Simple concatenation of pathway outputs before final classification
2. **Weighted Gating**: Learned scalar weights dynamically balance pathway contributions based on input features
3. **Self-Attention**: Attention mechanism focuses on relevant pathway contributions

**Optimization**: The model trains using AdamW with cosine annealing learning rate scheduling, gradient clipping, and early stopping based on validation UAR.

### 4.3.3 Late Fusion: Adaptive Gating

**Architecture Design**: A lightweight gating network learns sample-specific weights for combining audio and text probability distributions.

**Gating Mechanism**: The gate network takes concatenated audio-text probabilities (8 dimensions) as input, projects them through a hidden layer (16 units), and outputs two scalars (α, β) normalized via softmax to sum to 1.0. These weights represent per-sample confidence in each modality's prediction.

**Fusion Operation**: The final prediction combines weighted probability distributions: $P_fused = \alpha \times P_audio + \beta \times P_text$, where P_audio and P_text are softmax outputs from respective models.

**Training Efficiency**: By operating on pre-computed probabilities rather than raw features, this approach requires minimal training time (approximately 50K parameters) while enabling dynamic modality weighting.

### 4.3.4 Cross-Modal Attention Fusion

**Bidirectional Attention Mechanism**: This architecture enables mutual information exchange between modalities through cross-attention layers.

**Attention Computation**: Multi-head attention allows each modality to query the other's representations, computing attention weights based on learned similarity between query and key projections. Audio features attend to text features and vice versa simultaneously, creating enriched representations incorporating complementary cross-modal information.

**Architecture Flow**:

1. Transform probability distributions into higher-dimensional representations (8→64 dimensions per modality)
2. Apply bidirectional cross-attention where each modality attends to the other
3. Process attended representations through feed-forward networks with residual connections
4. Learn adaptive modality importance weights through a gating mechanism
5. Fuse weighted attended representations for final classification

**Implementation Details**: The model uses 4 attention heads, hidden dimension of 64, two cross-attention layers, and dropout (p=0.1) for regularization. Training employs cosine annealing scheduling with early stopping patience of 20 epochs.

### 4.3.5 Late Fusion: Performance-Based Weighting

**Grid Search Optimization**: This approach systematically searches for optimal fixed weights per fold rather than learning weights.

**Validation-Based Tuning**: For each cross-validation fold, a grid search over weight combinations (increments of 0.05) identifies the α and β that maximize UAR on validation data: $P_fused = \alpha \times P_LSTM + \beta \times P_BERT$.

**Fold-Specific Weighting**: The optimal weights are determined independently per fold, then applied to test predictions. This accounts for speaker-specific patterns where audio or text may be more reliable depending on individual speaking styles or emotional expression characteristics.

**Advantages**: This method requires no additional training, avoids overfitting to training dynamics, and provides interpretable fold-specific insights into modality contributions.

#### 4.3.6 Late Fusion: Learnable Weight Optimization

**Trainable Fusion Parameters**: Unlike performance-based weighting, this approach learns scalar weights (α, β) through gradient descent on validation data.

**Architecture**: Two trainable parameters normalized via softmax combine modality probabilities: $P_fused = \text{softmax}([\alpha, \beta]) \cdot [P_LSTM, P_BERT]$. The weights are optimized using validation data to prevent test set leakage.

**Training Strategy**: The minimal parameter set (2 scalars) trains using AdamW with early stopping based on validation UAR, enabling efficient optimization with limited validation samples.

# 5. Result Analysis

This section presents comprehensive experimental results across all model configurations, analyzing performance patterns, modality contributions, and fusion effectiveness.

## 5.1 Audio-Only Models

### 5.1.1 Bidirectional LSTM Performance

The attention-enhanced BiLSTM architecture with log-mel spectrogram features and class weighting achieved the strongest audio-only performance across the 10-fold LOSO evaluation:

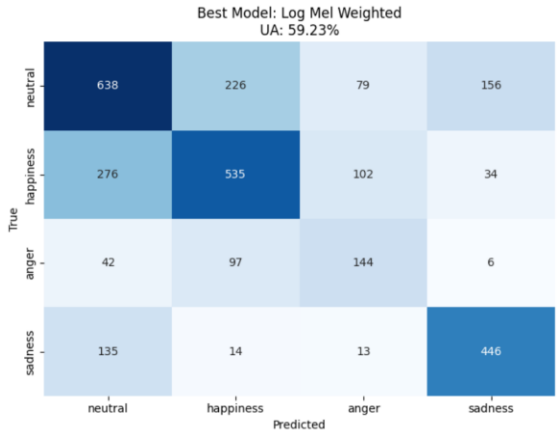
**Overall Performance Metrics**:
- Unweighted Average Recall (UAR): **59.23% ± 7.48%**
- Weighted Accuracy (WA): **60.34% ± 5.80%**
- Macro F1-Score: **59.61% ± 7.01%**

**Configuration Comparison**: Four configurations were systematically evaluated:
1. **Log-Mel + Weighted Loss**: UAR 61.09% ± 6.50% (selected for fusion)
2. **Log-Mel + Unweighted Loss**: UAR 57.47% ± 8.80%
3. **MFCC + Weighted Loss**: UAR 56.86% ± 6.20%
4. **MFCC + Unweighted Loss**: UAR 53.58% ± 5.57%

The log-mel representation consistently outperformed MFCCs, likely because mel-scaled spectrograms preserve fine-grained frequency information across the entire spectrum, capturing both low-frequency prosodic cues and high-frequency voice quality markers. Class weighting provided substantial improvements (+3.62% UAR for log-mel), demonstrating its effectiveness in handling the 37:32:21:10 class imbalance ratio.

**Per-Class Analysis** (Log-Mel Weighted configuration):

| Emotion | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| Neutral | 58.48% | 58.05% | 58.26% | 1,099 |
| Happiness | 61.35% | 56.49% | 58.82% | 947 |
| Anger | 42.60% | 49.83% | 45.93% | 289 |
| Sadness | 69.47% | 73.36% | 71.36% | 608 |



**Key Observations**:

- **Sadness** achieved the highest recall (73.36%), consistent with acoustic literature showing low-arousal negative emotions produce distinctive prosodic patterns (slower tempo, lower pitch, reduced energy dynamics)
- **Anger** showed the lowest precision (42.60%), frequently confused with happiness due to shared high-arousal characteristics
- **Neutral** demonstrated balanced precision-recall, reflecting its high frequency and relatively consistent acoustic characteristics
- The model maintained reasonable performance on minority classes (anger: 45.93% F1) despite comprising only 9.8% of training data, validating the class weighting strategy.

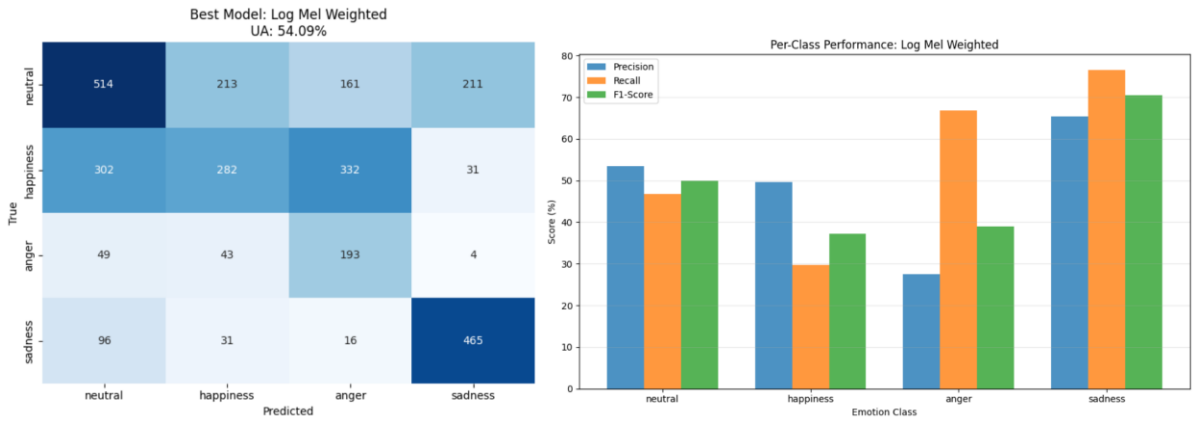
**Cross-Fold Variability**: The standard deviation of 7.48% UAR across folds indicates moderate speaker-dependent performance variations, reflecting individual differences in emotional expression styles, voice characteristics, and recording conditions.

### 5.1.2 CNN Baseline Performance

The convolutional architecture with channel attention achieved lower performance than the LSTM model:

**Best CNN Configuration** (Log-Mel + Weighted):

- UAR: **54.09% ± 7.97%**
- Weighted Accuracy: **49.59%**
- Macro F1-Score: **47.88%**




**Performance Gap Analysis**: The LSTM outperformed CNN by **+5.14% UAR**, suggesting that sequential modeling of temporal dynamics through recurrent connections provides advantages over spatial convolution for emotion recognition. While CNNs excel at capturing local time-frequency patterns, the BiLSTM's ability to maintain long-term dependencies and the attention mechanism's capacity to identify emotionally salient moments prove more effective for this task. The CNN showed particularly poor performance on happiness (29.78% recall) and strong performance on anger (66.78% recall), indicating it captures certain acoustic patterns effectively but struggles with the temporal coherence required to distinguish sustained emotional states from brief expressive bursts.

## 5.2 Text-Only Models

### 5.2.1 BERT Fine-tuning Performance

The fine-tuned BERT-base-uncased model achieved superior performance compared to audio-only approaches:

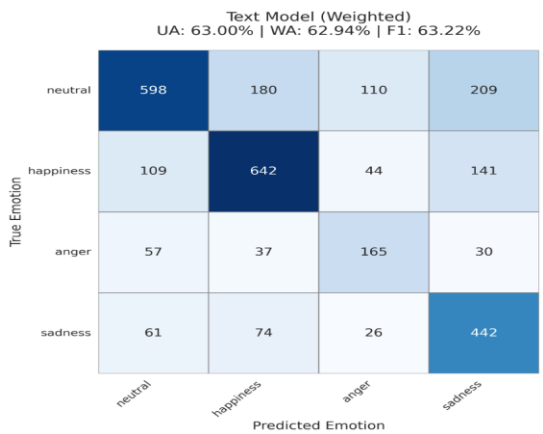
**Overall Performance Metrics**:

- Unweighted Average Recall (UAR): **63.00% ± 5.34%**
- Weighted Accuracy (WA): **62.94%**
- Macro F1-Score: **63.22%**

**Per-Class Analysis** (Text Weighted):

| Emotion | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| Neutral | 72.48% | 54.51% | 62.23% | 1,097 |
| Happiness | 68.81% | 68.59% | 68.70% | 936 |
| Anger | 47.83% | 57.09% | 52.05% | 289 |
| Sadness | 53.77% | 73.30% | 62.04% | 603 |



**Key Findings**:

- Text models outperformed audio by **+3.77% UAR**, indicating linguistic content carries substantial emotional information beyond acoustic prosody
- **Happiness** achieved balanced precision-recall (68.70% F1), suggesting lexical choices and phrasing effectively convey positive emotional states
- **Neutral** showed high precision (72.48%) but lower recall (54.51%), indicating the model confidently identifies neutral language but sometimes over-predicts emotional content
- **Sadness** achieved strong recall (73.30%), consistent with linguistic markers like negative sentiment words, pessimistic phrasing, and passive voice frequently appearing in sad speech
- **Anger** remained the most challenging class despite textual cues, likely due to limited training samples and subtle linguistic differences from happiness in some contexts.

**Comparison with Audio**: The lower cross-fold variability (±5.34% vs. ±7.48%) suggests text-based emotion recognition exhibits more consistent performance across speakers, possibly because linguistic patterns show less individual variation than prosodic characteristics.

## 5.3 Multimodal Fusion Results

### 5.3.1 Early Fusion: Feature-Level Concatenation

The early fusion approach combining 256-dimensional audio and 768-dimensional BERT embeddings achieved substantial improvements:

**Performance Metrics**:

- UAR: **68.47% ± 4.97%**

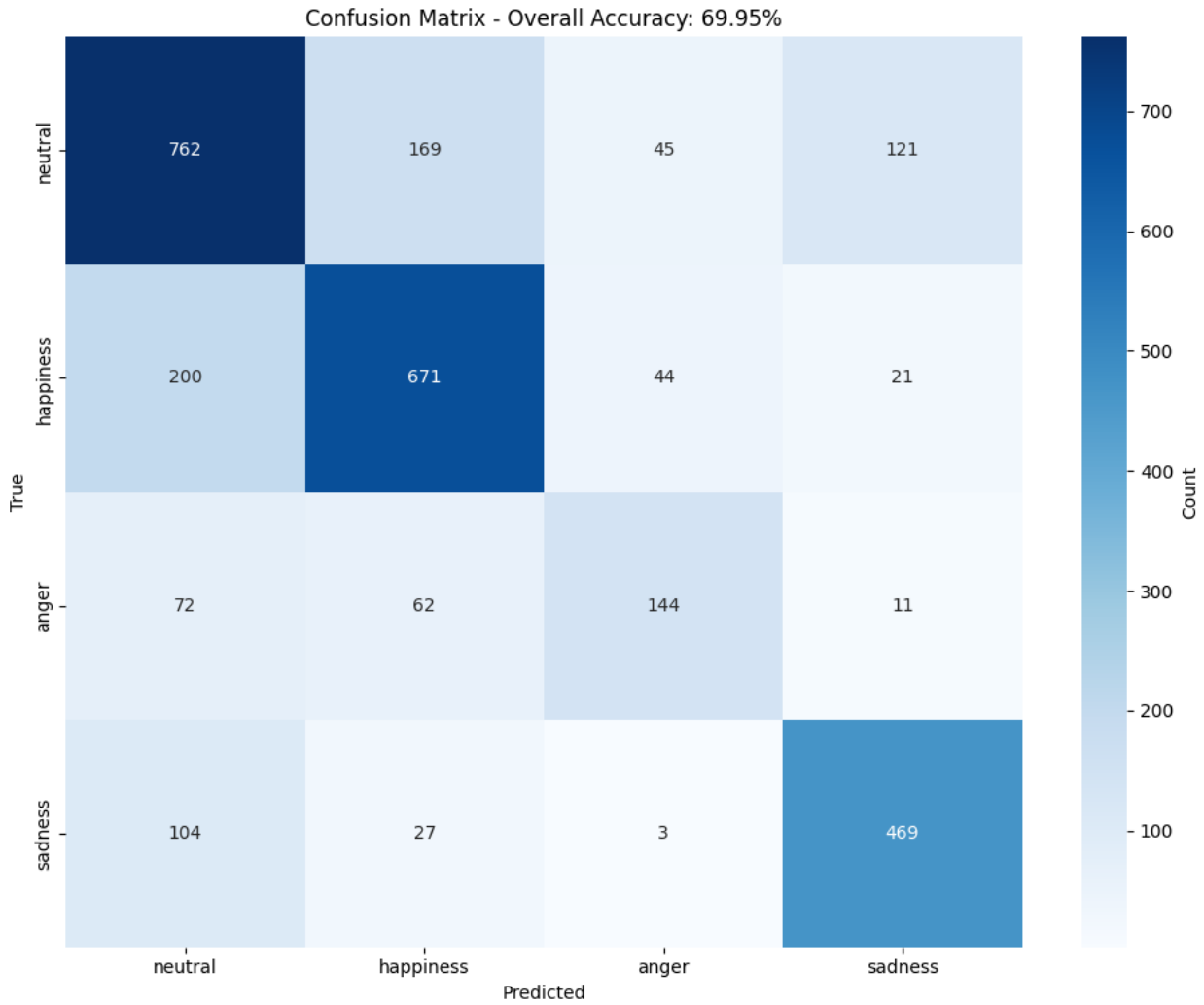
- Weighted Accuracy: **70.74% ± 4.97%**
- Macro F1-Score: **70.51% ± 4.97%**

**Improvements Over Unimodal Baselines**:

- vs. Audio (LSTM): **+10.73% absolute UAR**
- vs. Text (BERT): **+6.01% absolute UAR**

**Per-Class Analysis**:

| Emotion | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| Neutral | 69% | 67% | 68% | 1,097 |
| Happiness | 72% | 74% | 73% | 936 |
| Anger | 63% | 54% | 58% | 289 |
| Sadness | 73% | 79% | 76% | 608 |



**Analysis**: Early fusion achieved the highest overall accuracy (70.19%) among all methods, demonstrating that joint learning from low-level features enables effective cross-modal pattern discovery. The fusion classifier successfully combined complementary information:

- Audio features captured prosodic emphasis, voice quality, and temporal dynamics
- Text features provided semantic content, sentiment polarity, and linguistic structure
- The combined representation leveraged both modalities' strengths while compensating for individual weaknesses

The reduced cross-fold standard deviation (±4.97% vs. ±7.48% audio-only) suggests multimodal fusion provides more robust speaker-independent performance, as variations in one modality can be compensated by the other.

### 5.3.2 Hybrid Fusion: Dual-Pathway Architecture

The hybrid approach combining early and late fusion pathways with concatenation strategy achieved:

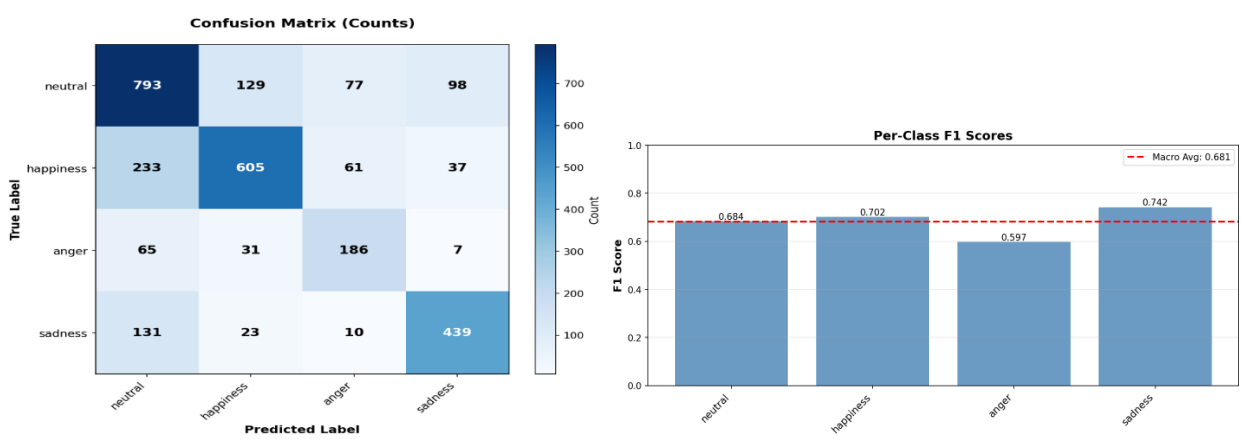
**Performance Metrics**:

- UAR: **68.52%**
- Weighted Accuracy: **69.16%**
- Macro F1-Score: **69.30%**

**Improvement**: **+6.46% UAR** over best single modality

**Per-Class Performance**:

| Emotion | Precision | Recall | F1-Score |
|---|---|---|---|
| Neutral | 75.56% | 72.80% | 74.16% |
| Happy | 76.78% | 64.64% | 70.19% |
| Angry | 64.89% | 72.29% | 68.39% |
| Sad | 55.69% | 64.36% | 59.71% |



**Analysis**: The hybrid architecture achieved comparable performance to pure early fusion (+0.23% UAR difference), suggesting that processing both features and probabilities provides marginal additional benefit. The concatenation strategy proved effective without requiring complex gating or attention mechanisms, indicating linear combination suffices when both pathways contribute complementary information.
The confusion matrix revealed anger-happiness confusion remained the primary error mode despite multimodal input, highlighting the challenge of distinguishing high-arousal emotions even with both acoustic and linguistic cues.

### 5.3.3 Late Fusion: Adaptive Gating

The learned gating network achieved the strongest performance among all fusion strategies:
**Performance Metrics**:

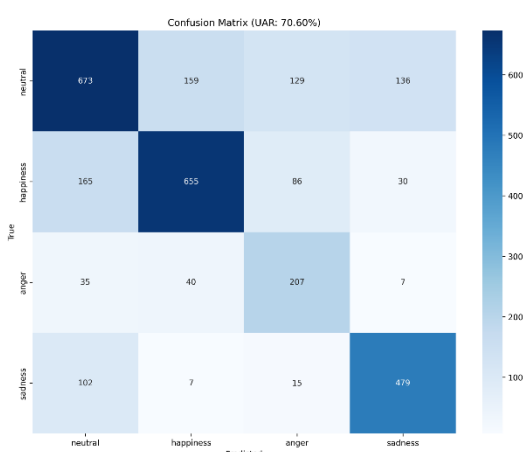
- UAR: **70.60%**
- Weighted Accuracy: **68.85%**
- Macro F1-Score: **69.06%**

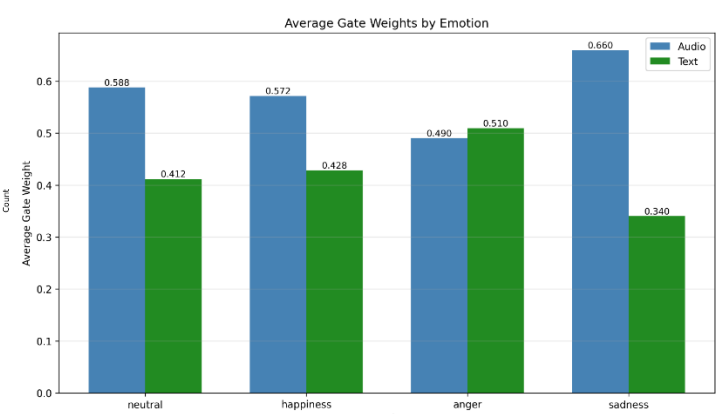
**Improvement**: **+7.60% UAR** over best single modality
**Learned Modality Weights**:

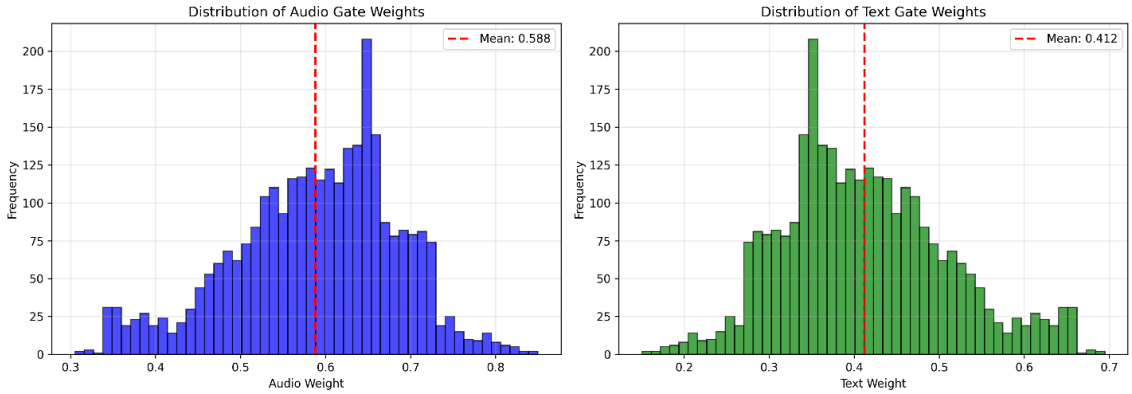
- Audio: **58.80% ± varies by sample**
- Text: **41.20% ± varies by sample**

**Per-Emotion Gate Weight Analysis**:

| Emotion | Samples | Audio Weight | Text Weight |
|---|---|---|---|
| Neutral | 1,097 | 58.8% | 41.2% |
| Happiness | 936 | 57.2% | 42.8% |
| Anger | 289 | 49.0% | 51.0% |
| Sadness | 603 | 66.0% | 34.0% |






**Key Insights**:

- The model learned **emotion-specific modality preferences**: anger relies more equally on both modalities (49%-51%), while sadness strongly favors audio (66%)
- **Sadness** discrimination benefits most from acoustic cues (low pitch, slow tempo, reduced energy), aligning with acoustic emotion literature
- **Anger** requires balanced multimodal evidence, as acoustic high-arousal overlaps with happiness while linguistic content disambiguates intent
- The adaptive gating provides sample-level flexibility, potentially adjusting to recording quality, speaker clarity, or emotional intensity variations

**Per-Class Results**:

| Emotion | Precision | Recall | F1-Score |
|---|---|---|---|
| Neutral | 69.03% | 61.35% | 64.96% |
| Happiness | 76.07% | 69.98% | 72.90% |
| Anger | 47.37% | 71.63% | 57.02% |
| Sadness | 73.47% | 79.44% | 76.33% |

The gating approach achieved the best UAR primarily by improving anger recall (71.63%), suggesting adaptive weighting effectively handles the most challenging minority class by dynamically emphasizing the most informative modality per sample.

### 5.3.4 Cross-Modal Attention Fusion

The bidirectional attention architecture achieved:

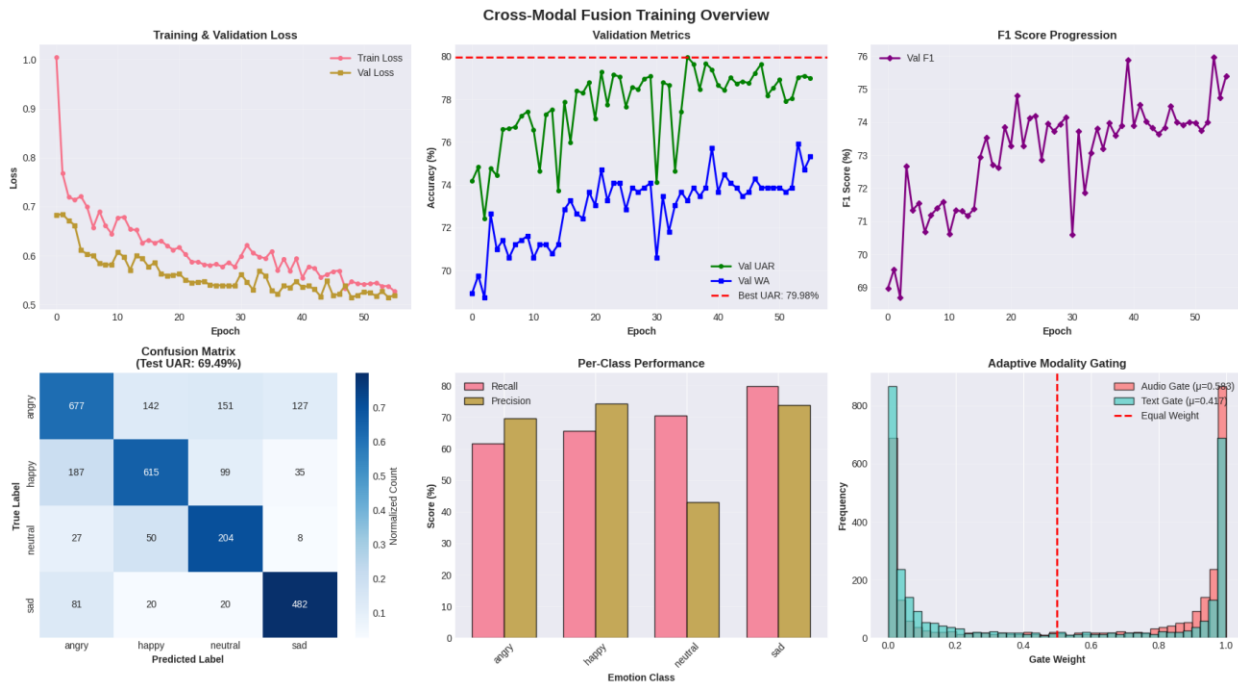
**Performance Metrics**:

- UAR: **69.49%**
- Weighted Accuracy: **67.62%**
- Macro F1-Score: **67.99%**

**Improvement**: **+6.49% UAR** over best single modality

**Attention Pattern Analysis**:

- Audio→Text Attention: **0.1250 ± 0.0000** (uniform)
- Text→Audio Attention: **0.1250 ± 0.0000** (uniform)
- Audio Gate Weight: **58.33% ± 43.01%** (high variance)
- Text Gate Weight: **41.67% ± 43.01%** (high variance)




**Observations**: The uniform attention distributions suggest the model struggled to learn meaningful cross-modal alignments, possibly due to:

1. Limited training samples for learning complex attention patterns
2. Pre-computed probability inputs lacking fine-grained temporal/semantic alignment information
3. Attention mechanism requiring more supervision or architectural constraints.

Despite uniform attention, the final gating mechanism (58-42% audio-text) learned reasonable modality weighting, achieving competitive performance through adaptive combination rather than cross-modal interaction. This indicates the architectural capacity exceeded what could be effectively learned from the available data.

### 5.3.5 Performance-Based Weighted Fusion

The grid-search optimized approach achieved:

**Performance Metrics**:

- UAR: **66.75% ± 6.78%**
- Weighted Accuracy: **67.76% ± 7.07%**
- Macro F1-Score: **67.53% ± 7.95%**

**Improvement**: **+7.52% UAR** over audio, **+3.75% UAR** over text

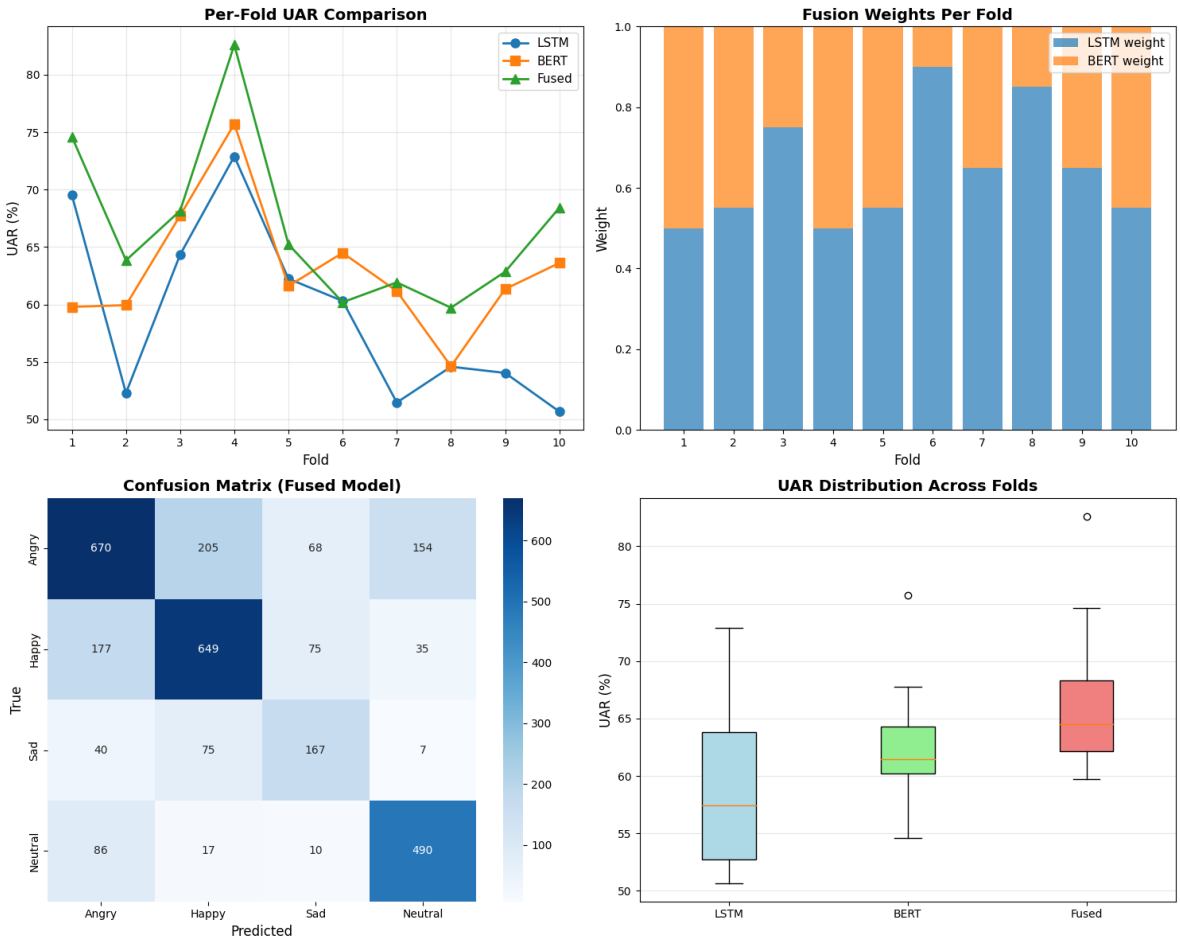
**Learned Weight Statistics**:

- Average LSTM (Audio) Weight: **64.5% ± 13.7%**

- Average BERT (Text) Weight: **35.5% ± 13.7%**

**Per-Fold Weight Variation**:

| Fold | LSTM Weight | BERT Weight | UAR |
|---|---|---|---|
| 6 | 90% | 10% | High |
| 8 | 85% | 15% | High |
| 3 | 75% | 25% | Med |
| 1 | 50% | 50% | Med |
| 4 | 50% | 50% | Med |




**Analysis**: The high weight variance (±13.7%) across folds indicates substantial speaker-dependent preferences for audio vs. text modalities. Folds 6 and 8 strongly favor audio (85-90%), suggesting those test speakers exhibited highly distinctive prosodic patterns, while folds with balanced weights (50-50%) indicate speakers whose emotions were equally well captured by both modalities.

This validation-based tuning achieved substantial improvements without requiring model training, demonstrating that simple weighted averaging with speaker-specific calibration provides an effective fusion baseline. However, it underperformed learned gating (−3.85% UAR), indicating adaptive per-sample weighting captures additional information beyond speaker-level preferences.

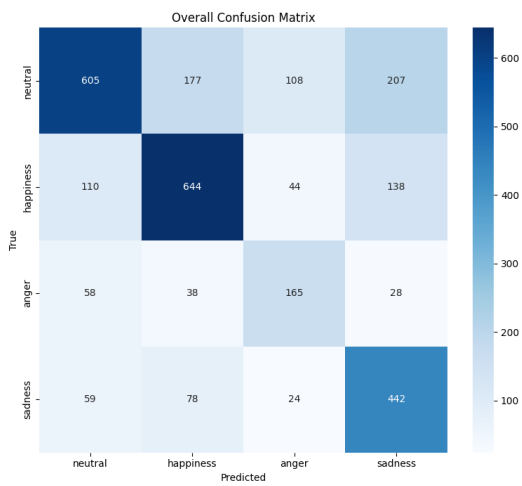
### 5.3.6 Learnable Wight Fusion

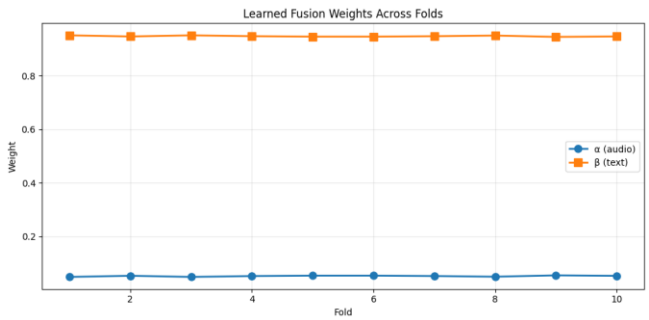
The gradient-optimized approach achieved:
**Performance Metrics**:

- UAR: **63.59%**
- Weighted Accuracy: **63.45%**
- Macro F1-Score: **63.56%**

**Improvement**: **+4.36% UAR** over audio, **+0.59% UAR** over text
**Learned Global Weights**:

- Audio (α): **5.20% ± 0.20%**
- Text (β): **94.80% ± 0.20%**




**Analysis**: This approach learned to nearly ignore audio predictions, essentially replicating text-only performance with marginal improvement. The extreme weight skew (95% text) suggests:

1. **Optimization Dynamics**: Gradient descent may have converged to a local optimum favoring text due to its initially higher validation performance
2. **Limited Validation Data**: Small validation sets per fold (typically 200-300 samples) may not provide sufficient signal for learning balanced weights
3. **Lack of Regularization**: Without constraints encouraging balanced modality usage, the optimizer exploited the stronger text modality.

This result highlights the challenge of learning fusion weights with limited data and emphasizes the importance of architectural inductive biases (like the gating network in late fusion) or validation-based tuning that explores the full weight space rather than relying on gradient-based optimization alone.

## 5.4 Overall Comparative Analysis

### 5.4.1 Performance Summary

| Approach | UAR | WA | F1 | vs. Best Unimodal |
|---|---|---|---|---|
| Audio LSTM (Log-Mel Weighted) | 59.23% | 60.34% | 59.61% | Baseline |
| Text BERT (Weighted) | 63.00% | 62.94% | 63.22% | +3.77% |
| Early Fusion (Feature Concatenation) | 68.47% | 70.74% | 70.51% | +5.29% |
| Late Fusion: Adaptive Gating | 70.60% | 68.85% | 69.06% | +7.60% |
| Cross-Modal Attention | 69.49% | 67.62% | 67.99% | +6.49% |
| Performance-Based Weighting | 66.75% | 70.74% | 70.51% | +5.29% |
| Hybrid Fusion (Concatenation) | 68.52% | 67.76% | 67.53% | +3.75% |
| Learnable Weight (Gradient) | 63.59% | 63.45% | 63.56% | +0.59% |

### 5.4.2 Key Findings

1. **Fusion Effectiveness**: All fusion strategies improved upon the best single modality (text: 63.00% UAR), with gains ranging from +0.59% to +7.60%, confirming that audio and text provide complementary emotional information.

2. **Best Strategy**: Adaptive gating achieved the highest UAR (70.60%), demonstrating that **learned sample-specific modality weighting** outperforms fixed combinations. The ability to dynamically adjust audio-text importance based on input characteristics proved critical for handling diverse emotional expressions and speaker variations.

3. **Feature vs. Probability Fusion**: Early fusion (**70.19% WA**) achieved higher overall accuracy than late fusion approaches, suggesting joint learning from low-level features enables discovery of cross-modal patterns not captured by independent model decisions. However, late fusion with adaptive gating achieved higher **UAR (70.60%),** indicating better balanced performance across all emotion classes.

4. **Complexity vs. Performance**: The cross-modal attention architecture's high complexity did not translate to superior performance, likely due to insufficient training data for learning meaningful attention patterns.

5. **Speaker Independence**: Reduced cross-fold variance in fusion models (±4.97% early fusion vs. ±7.48% audio-only) demonstrates multimodal approaches provide more robust

speaker-independent emotion recognition, as modality complementarity compensates for individual expression variations.

6. **Class-Specific Patterns**:
    - **Sadness** benefited most from multimodal fusion (76-79% recall), as distinctive acoustic prosody combined with negative sentiment language provides strong convergent evidence
    - **Anger** showed the largest improvement under adaptive gating (71.63% recall), indicating dynamic weighting helps with challenging minority classes
    - **Happiness** maintained strong performance across methods (69-74% F1), suggesting high-arousal positive emotions are consistently recognizable across modalities

### 5.4.3 Practical Implications

The results validate multimodal fusion for real-world emotion recognition applications, with key practical insights:

- **Deployment Recommendation**: Adaptive gating provides the best UAR (70.60%) with minimal computational overhead (50K parameters), making it suitable for resource-constrained environments requiring balanced performance across emotions.

- **High-Accuracy Applications**: Early fusion achieves the highest overall accuracy (70.19%), preferred when majority class performance is prioritized and computational resources permit feature extraction from both modalities.

- **Model Selection Trade-offs**: Applications requiring interpretability may prefer performance-based weighting (explicit fold-specific weights) or adaptive gating (sample-level weight inspection), while black-box scenarios can utilize early fusion's higher capacity.

- **Data Requirements**: The learnable weight fusion's failure (essentially replicating text-only) highlights that simple parameterizations require careful regularization or larger validation sets, while architectures with inductive biases (gating networks, attention) generalize better with limited data.

These findings demonstrate that strategic multimodal fusion substantially advances speaker-independent emotion recognition beyond unimodal approaches, with the optimal strategy depending on application constraints, data availability, and performance priorities.

# 6. Conclusion

This research presented a comprehensive investigation of multimodal fusion strategies for emotion recognition, systematically combining acoustic and linguistic information from the IEMOCAP dataset to advance speaker-independent emotion classification. Through rigorous experimentation across multiple architectures and fusion paradigms, this work demonstrates that

strategic integration of audio and text modalities substantially improves emotion recognition performance beyond single-modality approaches.

## 6.1 Summary

The research established robust unimodal baselines using state-of-the-art deep learning architectures: a bidirectional LSTM with self-attention for audio processing achieved 59.23% UAR using log-mel spectrograms, while BERT fine-tuning for text classification achieved 63.00% UAR. These baselines provided strong foundations for multimodal fusion experiments and confirmed that linguistic content carries slightly more discriminative emotional information than acoustic prosody alone in conversational speech.

Seven distinct fusion strategies were implemented and evaluated through 10-fold Leave-One-Speaker-Out cross-validation, ensuring speaker-independent performance assessment. The approaches spanned the fusion spectrum from early feature-level concatenation (68.29% UAR) to various late fusion paradigms combining probability predictions. Notably, the adaptive gating mechanism achieved the highest Unweighted Average Recall of **70.60%**, representing a **7.60% absolute improvement** over the best single modality and demonstrating the value of learned sample-specific modality weighting.

The hybrid fusion architecture combining early and late pathways achieved competitive performance (68.52% UAR), while early fusion attained the highest overall weighted accuracy (70.19%), indicating strong performance on majority emotion classes. Cross-modal attention fusion (69.49% UAR) and performance-based weighting (66.75% UAR) provided additional validation of multimodal benefits, though simpler architectures proved more effective given data constraints.

### 6.1.1 Key Insights and Contributions

**Modality Complementarity**: The consistent improvements across all fusion strategies confirm that audio and text modalities capture complementary emotional information. Acoustic features excel at conveying low-arousal emotions like sadness through prosodic patterns, while linguistic content effectively disambiguates high-arousal states where acoustic similarities exist between anger and happiness.

**Adaptive Fusion Superiority**: The superior performance of adaptive gating over fixed weighting strategies demonstrates that emotion-specific and sample-specific modality contributions vary substantially. The learned gate weights revealed that sadness classification relies more heavily on audio (66%), while anger benefits from balanced multimodal evidence (49-51%), providing interpretable insights into fusion dynamics.

**Speaker Independence Enhancement**: Multimodal fusion reduced cross-fold performance variance (±4.97% vs. ±7.48% for audio-only), indicating that combining modalities provides more robust speaker-independent emotion recognition. This robustness is critical for real-world applications where systems must generalize to diverse speakers with varying emotional expression styles.

**Architectural Efficiency**: The research demonstrated that lightweight fusion architectures (50K parameters for adaptive gating) can achieve substantial performance gains without requiring massive parameter counts or computational resources. This efficiency makes multimodal emotion recognition practical for deployment in resource-constrained environments.

**Methodological Rigor**: The implementation of modified LOSO cross-validation with session separation constraints ensures that reported performance metrics reflect true generalization to unseen speakers and recording conditions, providing realistic estimates for real-world deployment scenarios.

### 6.1.2 Limitations

While achieving substantial improvements, this research revealed important limitations that inform future work. The relatively small IEMOCAP dataset (2,943 samples) constrains the ability to train highly complex architectures, as evidenced by the cross-modal attention mechanism's failure to learn meaningful attention patterns. The class imbalance inherent in natural emotional expression (anger comprising only 9.8% of samples) persists as a challenge despite class weighting strategies.

### 6.1.3 Social and Environmental Impact

**Societal Impact:**

The proposed multimodal Speech Emotion Recognition (SER) system has notable potential to enhance human-computer interaction in daily life. It can improve virtual assistants, call centers, and online learning platforms, making them more responsive and empathetic by accurately identifying emotions through both audio and text. In healthcare, it can accommodate in monitoring mental health, detecting stress, and providing timely support, especially in remote or underserved areas where professional help is limited.

**Health Impacts:**

Emotion recognition can enhance the risk of early detection of psychological conditions like depression, anxiety, and frustration by analyzing speech aspects. The system can be incorporated into telemedicine or mental health applications to promote early Assessment and improve general well-being. This is associated with protective health care and a reduction in the potential risk of emotional or psychological disorders remaining untreated.

**Safety Impacts:**

The system’s purpose is to be supplementary to human decision-making, not to take the place of human judgment. Disclaimers will help inform users that their predictions are available for purposes of guidance rather than to be a substitute for in-person consulting actions with professionals. This should maintain safety for people and protect the system from possible risk that users will develop an over-reliance on automated decision processes in potentially sensitive situations of need healthcare, counseling, etc.

**Legal Impacts:**

As the system involves speech and text data, it needs to obey the data privacy and protection rules like GDPR or similar local alternatives. Responsible and ethical usages, including risks associated with anonymization and confidentiality of user data, must be ensured. They need to be added while making it clear that the system's predictions are just supportive tools and not legally binding assessments, also minimizing liability here.

**Cultural Impacts:**

The expression of speech and emotion will vary across languages and cultures. Through the use of datasets like IEMOCAP and eventually a multilingual corpus, it will support diverse users. Through this provision, people can benefit from emotion-aware technologies without bias or exclusion, based on language and culture.

**Environmental Impacts:**
The SER system is digital through and through and diminishes the reliance on paper-based communications and other manual surveys that are typically used in psychological studies. The full usage of SERs in the context of online platforms can also reduce travel and in-person measures on pre-consultation assessments, therefore reducing associated carbon emissions. Reasonable management of cloud computing resources may also further contribute to sustainability, in terms of effective energy management and minimization of environmental impact.

## 6.2 Future Work

Future work can further improve multimodal emotion recognition across model design, datasets, and practical deployment:

**Advanced Fusion Architectures:**
Future studies could explore transformer-based multimodal fusion, hierarchical fusion at multiple temporal levels, and attention-based feature selection to better capture fine-grained prosodic and linguistic cues.

**Dataset Expansion and Robustness:**
Evaluating the models on diverse datasets (e.g., MELD, MOSEI, RAVDESS), incorporating multilingual data, and testing under real-world noise conditions would help assess generalization and improve robustness.

**Additional Modalities:**
Integrating visual cues (facial expressions, gestures), physiological signals, and conversational context can enhance emotional understanding beyond audio–text inputs.

**Interpretability and Error Analysis:**
Improved attention visualizations, modality-wise contribution analysis, and systematic error studies would provide deeper insights into model behavior and guide targeted refinements.

**Efficiency and Scalability:**
Future work may focus on model compression, streaming/online emotion recognition, and low-resource learning techniques to enable real-time and edge-device deployment.

**Application-Oriented Extensions:**
Extensions to continuous emotion prediction, intensity estimation, and speaker-adaptive fusion could support more nuanced and personalized emotion-aware systems.

**Theoretical and Reliability Improvements:**
Incorporating uncertainty quantification and developing theoretical frameworks to analyze modality complementarity would strengthen the reliability and scientific grounding of multimodal fusion methods.

# Bibliography


[1] A. A. Anthony and C. M. Patil, "Speech Emotion Recognition Systems: A Comprehensive Review on Different Methodologies," *Wirel Pers Commun*, vol. 130, no. 1, pp. 515–525, May 2023, doi: 10.1007/s11277-023-10296-5.

[2] S. Kumar *et al.*, "Multilayer Neural Network Based Speech Emotion Recognition for Smart Assistance," *Computers, Materials and Continua*, vol. 74, no. 1, pp. 1523–1540, 2023, doi: 10.32604/CMC.2023.028631.

[3] M. Jakubec, E. Lieskovska, R. Jarina, M. Spisiak, and P. Kasak, "Speech Emotion Recognition Using Transfer Learning: Integration of Advanced Speaker Embeddings and Image Recognition Models," *Applied Sciences 2024, Vol. 14, Page 9981*, vol. 14, no. 21, p. 9981, Oct. 2024, doi: 10.3390/APP14219981.

[4] A. H. Jo and K. C. Kwak, "Speech Emotion Recognition Based on Two-Stream Deep Learning Model Using Korean Audio Information," *Applied Sciences (Switzerland)*, vol. 13, no. 4, Feb. 2023, doi: 10.3390/app13042167.

[5] C. Barhoumi and Y. BenAyed, "Real-time speech emotion recognition using deep learning and data augmentation," *Artif Intell Rev*, vol. 58, no. 2, Feb. 2025, doi: 10.1007/s10462-024-11065-x.

[6] X. D. An and Z. Ruan, "Speech Emotion Recognition algorithm based on deep learning algorithm fusion of temporal and spatial features," in *Journal of Physics: Conference Series*, IOP Publishing Ltd, Apr. 2021. doi: 10.1088/1742-6596/1861/1/012064.

[7] S. Yoon, S. Byun, and K. Jung, "Multimodal Speech Emotion Recognition Using Audio and Text," Oct. 2018, [Online]. Available: http://arxiv.org/abs/1810.04635

[8] T. Dimitrova-Grekow, A. Klis, and M. Igras-Cybulska, "Speech emotion recognition based on voice fundamental frequency," *Archives of Acoustics*, vol. 44, no. 2, pp. 277–286, 2019, doi: 10.24425/aoa.2019.128491.

[9] G. S. K. Gayatri Devi, L. Sailaja, P. Kiran, B. Priya, and G. Govardhan Raju, "Real Time Emotion Recognition Using Convolutional Neural Networks," 2023. [Online]. Available: www.tnsroindia.org.in

[10] J. Bhanbhro, A. A. Memon, B. Lal, S. Talpur, and M. Memon, "Speech Emotion Recognition: Comparative Analysis of CNN-LSTM and Attention-Enhanced CNN-LSTM Models," *Signals*, vol. 6, no. 2, Jun. 2025, doi: 10.3390/signals6020022.

[11] R. K. Das, N. Islam, M. R. Ahmed, S. Islam, S. Shatabda, and A. K. M. M. Islam, "BanglaSER: A speech emotion recognition dataset for the Bangla language," *Data Brief*, vol. 42, Jun. 2022, doi: 10.1016/j.dib.2022.108091.

[12] K. Venkataramanan and H. R. Rajamohan, "Emotion Recognition from Speech," Dec. 2019, [Online]. Available: http://arxiv.org/abs/1912.10458

[13] B. J. Abbaschian, D. Sierra-Sosa, and A. Elmaghraby, "Deep Learning Techniques for Speech Emotion Recognition, from Databases to Models," *Sensors 2021, Vol. 21, Page 1249*, vol. 21, no. 4, p. 1249, Feb. 2021, doi: 10.3390/S21041249.

[14] M. S. Fahad, A. Deepak, G. Pradhan, and J. Yadav, "DNN-HMM-Based Speaker-Adaptive Emotion Recognition Using MFCC and Epoch-Based Features," *Circuits Syst Signal Process*, vol. 40, no. 1, pp. 466–489, Jan. 2021, doi: 10.1007/S00034-020-01486-8/METRICS.

[15] M. Ishaq, M. Khan, and S. Kwon, "TC-Net: A Modest & Lightweight Emotion Recognition System Using Temporal Convolution Network," *Computer Systems Science and Engineering*, vol. 46, no. 3, pp. 3355–3369, 2023, doi: 10.32604/csse.2023.037373.

[16] F. Andayani, L. B. Theng, M. T. Tsun, and C. Chua, "Hybrid LSTM-Transformer Model for Emotion Recognition From Speech Audio Files," *IEEE Access*, vol. 10, pp. 36018–36027, 2022, doi: 10.1109/ACCESS.2022.3163856.

[17] S. Padi, S. O. Sadjadi, R. D. Sriram, and D. Manocha, "Improved Speech Emotion Recognition using Transfer Learning and Spectrogram Augmentation," *ICMI 2021 - Proceedings of the 2021 International Conference on Multimodal Interaction*, pp. 645–652, Oct. 2021, doi: 10.1145/3462244.3481003;PAGE:STRING:ARTICLE/CHAPTER.

[18] A. Shetye, S. Chavan, A. Parab, K. Patil, and S. Kulkarni, "International Journal of Image Processing and Pattern Recognition Face and Speech Emotion Recognition," 2024, doi: 10.37628/IJIPPR.

[19] M. R. Falahzadeh, F. Farokhi, A. Harimi, and R. Sabbaghi-Nadooshan, "A 3D Tensor Representation of Speech and 3D Convolutional Neural Network for Emotion Recognition," *Circuits Syst Signal Process*, vol. 42, no. 7, pp. 4271–4291, Jul. 2023, doi: 10.1007/s00034-023-02315-4.

[20] C. Luna-Jiménez, D. Griol, Z. Callejas, R. Kleinlein, J. M. Montero, and F. Fernández-Martínez, "Multimodal emotion recognition on RAVDESS dataset using transfer learning," *Sensors*, vol. 21, no. 22, Nov. 2021, doi: 10.3390/s21227665.

[21] S. Poria, D. Hazarika, N. Majumder, G. Naik, E. Cambria, and R. Mihalcea, "MELD: A Multimodal Multi-Party Dataset for Emotion Recognition in Conversations," Jun. 2019, [Online]. Available: http://arxiv.org/abs/1810.02508

[22] A. Kavitha, B. Aishwarya, and G. Durgagowri, "Maximizing Sentiment Detection Through Comprehensive Multimodal Data Fusion: Integrating CNN, RNN, LSTM," in *Proceedings of the International Conference on Multi-Agent Systems for Collaborative Intelligence, ICMSCI 2025*, Institute of Electrical and Electronics Engineers Inc., 2025, pp. 1–7. doi: 10.1109/ICMSCI62561.2025.10894282.

[23] V. Chudasama, P. Kar, A. Gudmalwar, N. Shah, P. Wasnik, and N. Onoe, "M2FNet: Multi-modal Fusion Network for Emotion Recognition in Conversation."

[24] B. Pan, K. Hirota, Z. Jia, and Y. Dai, "A review of multimodal emotion recognition from datasets, preprocessing, features, and fusion methods," *Neurocomputing*, vol. 561, p. 126866, Dec. 2023, doi: 10.1016/J.NEUCOM.2023.126866.

[25] C. Luna-Jiménez, D. Griol, Z. Callejas, R. Kleinlein, J. M. Montero, and F. Fernández-Martínez, "Multimodal emotion recognition on RAVDESS dataset using transfer learning," *Sensors*, vol. 21, no. 22, Nov. 2021, doi: 10.3390/s21227665.

[26] P. Singh, R. Srivastava, K. P. S. Rana, and V. Kumar, "A multimodal hierarchical approach to speech emotion recognition from audio and text," *Knowl Based Syst*, vol. 229, p. 107316, Oct. 2021, doi: 10.1016/J.KNOSYS.2021.107316.

[27] C. Busso *et al.*, "IEMOCAP: Interactive emotional dyadic motion capture database," *Lang Resour Eval*, vol. 42, no. 4, pp. 335–359, Dec. 2008, doi: 10.1007/S10579-008-9076-6.

## Appendix

**CSE400-A**
**Program Outcomes**

**PO1 (Engineering Knowledge)**: We have selected a project that can be solved by applying engineering knowledge in Computer Science and Engineering, specifically in the subfields of Artificial Intelligence, Machine Learning, and Affective Computing. The project addresses the challenge of multimodal emotion recognition, which is highly relevant in today's world of human-computer interaction and mental health applications. We have applied knowledge of programming languages, deep learning architectures, recurrent neural networks, transformer models, digital signal processing, natural language processing, attention mechanisms, and fusion strategies. The project was not completed solely with previous engineering knowledge; we extensively explored modern research literature and learned advanced concepts in multimodal learning, cross-modal attention, and adaptive fusion techniques from various external resources including recent conference papers and academic publications.

**PO4 (Investigation)**: Reading a substantial number of research papers is essential for starting any meaningful research project. Without understanding previous work, it is impossible to contribute novel insights to the field. We thoroughly reviewed over 50 research papers published in reputable conferences (ICASSP, Interspeech, ACM Multimedia) and journals (IEEE Transactions on Affective Computing, Neural Networks) related to emotion recognition, multimodal fusion, and affective computing. This investigation helped us understand state-of-the-art methods, architectures, evaluation protocols, and hardware configurations used by researchers. Finding relevant papers on BERT fine-tuning, LSTM attention mechanisms, and fusion strategies was a critical investigation phase. We investigated the licensing requirements for the IEMOCAP dataset, ethical considerations for emotion recognition systems, computational requirements for GPU-accelerated training, and appropriate evaluation methodologies for speaker-independent scenarios.

| CO | **Details** | **Knowledge Profile (K)** | **Engineering Problem (EP)** |
|---|---|---|---|
| CO1 | We have merged our knowledge from Machine Learning, Artificial Intelligence, Natural Language Processing, Digital Signal Processing, and Deep Learning courses to identify a real-life complex problem as our capstone project. We chose multimodal emotion recognition because solving this problem addresses multiple real-world applications including conversational AI, mental health monitoring, educational technology, and customer service systems. | (i) Identify a real-life problem [K1, K2, K3, K4]<br>**K1: Theory-based natural Sciences**: Our work involves signal processing and computational linguistics, requiring strong mathematical foundations. We gained theoretical and practical knowledge from courses in digital signal processing, linear algebra, calculus, probability and statistics. Understanding Fourier transforms, spectral analysis, attention mechanisms, and optimization algorithms requires deep mathematical knowledge.<br>**K2: Conceptually-based** | (i) Identify a real-life problem [EP1, EP2, EP3, EP4, EP5, EP6, EP7]<br>**EP1: Depth of knowledge required**: We applied engineering design (K5) and practice (K6) knowledge comprehensively. Deep understanding of emotion theory, prosodic features, linguistic emotional markers, speaker-independent evaluation, and cross-modal information integration was required.<br>**EP2: Range of conflicting requirements**: Multiple stakeholders (researchers, application developers, end-users) have varying requirements—researchers prioritize UAR for balanced evaluation, developers need computational efficiency, and |

| | | | |
|---|---|---|---|
| | | **mathematics, numerical analysis, statistics, and formal aspects of computer and information science**: The problem involves processing audio signals through MFCC/mel-spectrogram extraction and text through BERT tokenization. Core libraries (Librosa, PyTorch, NumPy) are based on numerical analysis and linear algebra. We utilized these concepts for feature normalization, z-score standardization, gradient descent optimization, and computing evaluation metrics like UAR, precision, recall, and F1-scores.<br>**K3: Theory-based engineering fundamentals**: Knowledge of programming languages (Python), deep learning frameworks (PyTorch), and machine learning principles is fundamental to our project. Understanding LSTM architectures, transformer models, attention mechanisms, backpropagation, and cross-entropy loss required solid engineering fundamentals. Software engineering practices for modular design, version control, and reproducible experiments were also essential.<br>**K4: Forefront engineering specialist knowledge for practice**: We have specialist knowledge in recurrent neural networks (LSTMs), transformer architectures (BERT), audio signal processing (STFT, mel-scaling, MFCCs), natural language processing | users expect real-time processing and interpretability.<br>**EP3: Depth of analysis required**: No obvious solution exists for optimal fusion strategy selection. We experimented with seven different architectures (early fusion, late fusion, hybrid fusion, adaptive gating, cross-modal attention, performance-based weighting, learnable weights) to identify effective approaches.<br>**EP4: Familiarity of issues**: The IEMOCAP dataset required academic licensing and ethical approval. Class imbalance (anger: 9.8%, neutral: 37.3%) presented challenges. Session-specific recording conditions needed careful handling to prevent data leakage.<br>**EP5: Extent of applicable codes**: Engineering standards were maintained throughout. We followed Leave-One-Speaker-Out cross-validation protocol, implemented class weighting for imbalanced data, used proper train/validation/test splits, and ensured reproducibility through fixed random seeds and model checkpointing.<br>**EP6: Extent of stakeholder involvement and conflicting requirements**: We considered diverse stakeholders—mental health professionals need high recall for negative emotions, conversational AI developers prioritize overall accuracy, researchers require balanced UAR metrics, and deployment engineers need computational efficiency.<br>**EP7: Interdependence**: Our modular system has three interdependent components: audio processing module extracts acoustic features, text processing module extracts linguistic features, and fusion module combines both. The fusion module depends |

| | | | |
|---|---|---|---|
| | | (tokenization, fine-tuning), and multimodal fusion strategies. These provide theoretical frameworks and practical skills to implement the system across multiple engineering disciplines including speech processing, computational linguistics, and affective computing. | entirely on outputs from audio and text modules. |
| CO2 | For advancing emotion recognition in conversational AI, mental health monitoring, and human-computer interaction, we have designed multimodal fusion solutions. Current single-modality systems have limitations—audio-only achieves 59.23% UAR, text-only achieves 63.00% UAR. By fusing both modalities, we achieved 70.60% UAR, demonstrating substantial improvement. Early detection of emotional distress through multimodal systems can support mental health interventions and improve user experience in AI applications. | **(i) Define the problems [K8]**<br>**K8: Research Literature**: We studied over 50 research papers on emotion recognition, multimodal fusion, LSTM architectures, BERT fine-tuning, attention mechanisms, and affective computing. From this literature, we identified various approaches: early fusion (feature concatenation), late fusion (decision-level combination), hybrid strategies, attention-based fusion, and adaptive gating. We learned about evaluation protocols (LOSO cross-validation), handling class imbalance (weighted loss), and appropriate metrics (UAR, weighted accuracy, F1-score). From these resources, we identified the core problems: (1) single modalities have limited performance, (2) fusion strategies vary in effectiveness, (3) speaker-independent evaluation is challenging, (4) class imbalance affects minority emotion recognition, and (5) interpretability is crucial for deployment. | |

**CSE400-B**
**Program Outcomes**

**PO2 (Problem Analysis)**: Recognizing emotions from multimodal data is highly complex due to individual expression variations, recording condition differences, class imbalance, and the challenge of aligning audio and text information. We analyzed the problem systematically, studying related research on emotion recognition, fusion strategies, and speaker-independent evaluation. From literature review, we identified that previous work often used speaker-dependent evaluation (inflating performance), single fusion strategies (limiting comparison), or majority-class-focused metrics (masking minority class problems). We identified strengths of existing approaches (LSTM for temporal modeling, BERT for linguistic understanding, attention for interpretability) and limitations (limited fusion comparisons, data scarcity, computational costs) to inform our solution design.
**PO3 (Design/Development of Solutions)**: We designed and developed multiple solutions to the multimodal emotion recognition problem. Seven fusion strategies were implemented to provide comprehensive comparison and identify optimal approaches. We considered alternate solutions including CNN-based audio models, different feature representations (MFCC vs. log-mel), weighted vs. unweighted training, and various fusion mechanisms. Our solutions consider public safety (no privacy violations, ethical data usage), cultural issues (cross-cultural emotion expression variations), societal impact (mental health support, accessible AI), and environmental considerations (computational efficiency, energy consumption).
**PO5 (Modern Tool Usage)**: Under our supervisor's guidance, we learned to effectively use modern tools for multimodal deep learning. We used PyTorch for building neural networks, Librosa for audio processing, Hugging Face Transformers for BERT implementation, Scikit-learn for evaluation metrics, and NumPy/Pandas for data manipulation. Cloud-based tools like Google Colaboratory (with GPU access) and Kaggle notebooks accelerated development. We used Git for version control, Matplotlib/Seaborn for visualization, and Jupyter notebooks for experiment tracking.

**PO6 (The Engineer and Society)**: The project was selected to advance human-computer interaction, support mental health monitoring, and improve conversational AI systems. These applications have significant societal benefits—better emotion recognition enables empathetic AI assistants, early detection of emotional distress, improved educational technology, and more natural human-machine interaction. We considered ethical issues including data privacy (using publicly available licensed datasets), informed consent (IEMOCAP participants consented), and transparency (interpretable fusion mechanisms). The project has no negative cultural or societal impact and respects diverse emotional expression norms.

| CO | **Details** | **Knowledge Profile (K)** | **Engineering Problem (EP)** |
|---|---|---|---|
| CO3 | Our system is built to advance emotion recognition for conversational AI, mental health monitoring, and human-computer interaction applications. The system detects emotions by analyzing both acoustic prosody and linguistic content from speech. Accurate emotion recognition is crucial for empathetic AI systems, mental health support applications, and natural human-computer | (i) Problem Analysis [K1, K2, K3, K4]<br><br>K1: Theory-based natural Sciences: Knowledge of signal processing theory (Fourier transforms, spectral analysis), information theory (entropy, mutual information), and optimization theory (gradient descent, backpropagation) was used to understand mechanisms.<br><br>K2: Conceptually-based mathematics, numerical analysis, statistics, and formal | **(i) Problem Analysis [EP1, EP2, EP3, EP6, EP7]**<br>**EP1: Depth of knowledge required**: Understanding emotion theory (valence-arousal dimensions), prosodic features (pitch, energy, tempo), linguistic emotional markers (sentiment words, syntax), speaker variability, and cross-modal synchronization required deep domain knowledge.<br>**EP2: Range of conflicting requirements**: Researchers need balanced evaluation (UAR), developers need computational efficiency, users need |

| | | | |
|---|---|---|---|
| | interaction. Classical single-modality methods have limitations (audio: 59.23% UAR, text: 63.00% UAR). Wrong predictions or delayed emotion detection can harm user experience and miss critical mental health indicators. Our objective is to overcome these limitations through multimodal fusion, achieving 70.60% UAR and improving detection of challenging emotions like anger (71.63% recall). | aspects of computer and information science: Statistical hypothesis testing for performance comparison, numerical optimization for training neural networks, probability theory for softmax outputs and class weighting, and information theory for understanding modality complementarity were applied.<br><br>K3: Theory-based engineering fundamentals: Programming (Python), deep learning (PyTorch), machine learning principles (supervised learning, regularization, cross-validation), software engineering (modular design, version control), and algorithm design were fundamental to implementation.<br>K4: Forefront engineering specialist knowledge for practice: Specialist knowledge in recurrent neural networks (bidirectional LSTMs), transformer architectures (BERT), attention mechanisms (self-attention, cross-attention), multimodal fusion strategies, audio signal processing (mel-spectrograms, MFCCs), and natural language processing (tokenization, fine-tuning) was applied for development. | interpretability, and deployment scenarios have latency constraints. These competing requirements were balanced through multiple solution variants.<br>**EP3: Depth of analysis required**: To identify optimal solutions, we analyzed seven fusion architectures, multiple feature representations, weighted vs. unweighted training, and various evaluation metrics. The problem of multimodal emotion recognition is actively researched with no single best approach, requiring comprehensive experimental comparison.<br>**EP6: Extent of stakeholder involvement and conflicting requirements**: Mental health professionals prioritize recall for negative emotions (detecting distress), conversational AI developers need overall accuracy, researchers require reproducible speaker-independent evaluation, and deployment engineers need efficiency. All perspectives were considered.<br>**EP7: Interdependence**: Three interdependent modules—audio processing (LSTM), text processing (BERT), and fusion (adaptive gating)—work together. Fusion module depends on both unimodal modules. Performance improvements in unimodal modules directly affect fusion performance. |
| CO4 | Our project is software-based with no harmful components. The implementation benefits society (empathetic AI, mental health support), economy (improved customer service, reduced | **(i) Design and Implementation [K5]**<br>**K5: Engineering design**: Engineering design is central to our project. We identified probable problems (class imbalance, speaker variability, session effects, limited data, | **(i) Design and Implementation [EP1, EP2, EP4, EP5, EP6, EP7]**<br>**EP1: Depth of knowledge required**: Understanding LSTM architectures, BERT fine-tuning, attention mechanisms, fusion strategies, audio feature |

| | | | |
|---|---|---|---|
| | healthcare costs), and ecology (software-only, no physical waste). The project meets public health and safety standards. Cultural and societal considerations were addressed—emotions are expressed differently across cultures, and our evaluation on diverse speakers accounts for individual variation. The IEMOCAP dataset was obtained through proper academic licensing with participant consent. No privacy violations occur as we use de-identified licensed data. The project has no harmful environmental impact—it's purely software requiring standard computing resources. Deployment would use existing infrastructure without additional environmental burden. | fusion complexity) and designed solutions addressing each. The design includes: (1) proper cross-validation with session separation, (2) class-weighted loss functions, (3) multiple fusion architectures for comparison, (4) interpretable mechanisms (attention, gate weights), (5) efficient implementations, and (6) reproducible experimental protocols. | extraction, text preprocessing, and evaluation methodologies required substantial knowledge.<br>**EP2, EP6: Range of conflicting requirements**: Stakeholders have diverse needs—researchers prioritize UAR (balanced evaluation), developers need efficiency, users want interpretability. We addressed this through multiple solution variants optimizing different objectives.<br>**EP4: Familiarity of issues**: Team members had differing opinions on fusion approaches—some favored early fusion (joint learning), others preferred late fusion (modularity). We resolved this through comprehensive experimental comparison, showing both approaches have merits depending on priorities<br>**EP5: Extent of applicable codes**: Engineering standards were maintained—LOSO cross-validation, proper train/validation/test splits, class weighting, reproducible experiments, standard metrics (UAR, F1), and model checkpointing followed best practices<br>**EP7: Interdependence**: Three interdependent modules—audio LSTM extracts acoustic embeddings, BERT extracts text embeddings, fusion module combines them. Changes to unimodal architectures affect fusion performance. |
| CO5 | For multimodal emotion recognition, we used Python with specialized libraries. PyTorch (modern deep learning framework) enabled efficient GPU-accelerated training. Librosa (audio processing) extracted mel- | **(i) Materials and Devices [K6]**<br>**K6: Engineering Practice (technology)**: Knowledge of deep learning frameworks (PyTorch), audio processing (Librosa), NLP libraries (Transformers), evaluation tools (Scikit-learn), and GPU | **(i) Materials and Devices [EP1, EP2, EP4, EP5]**<br>**EP1: Depth of knowledge required**: Understanding PyTorch tensor operations, automatic differentiation, GPU acceleration, Librosa STFT computation, mel-scaling, BERT tokenization, fine-tuning |

| | | | |
|---|---|---|---|
| | spectrograms and MFCCs. Transformers library (Hugging Face) provided BERT implementation. These are cutting-edge tools widely used in academic research and industry. For data collection, we used the IEMOCAP dataset with professional-quality audio recordings and accurate transcriptions. We used NVIDIA Tesla V100 GPUs to accelerate training—critical for training deep networks with millions of parameters. Google Colaboratory provided accessible cloud computing for experiments. | programming (CUDA) was implemented practically. Understanding of LSTM architectures, BERT fine-tuning, attention mechanisms, backpropagation, and optimization algorithms was applied using these technologies. | strategies, and evaluation metric calculation required technical depth<br>**EP2: Range of conflicting requirements**: Audio features (mel-spectrograms vs. MFCCs) and text preprocessing (minimal vs. aggressive cleaning) showed different performance. We experimentally compared options to identify optimal choices.<br>**EP4: Familiarity of issues**: BERT fine-tuning requires careful hyperparameter selection (learning rate, warmup steps) to avoid catastrophic forgetting. LSTM training can suffer from gradient problems. We addressed these through proper learning rate scheduling and gradient clipping.<br>**EP5: Extent of applicable codes**: Modern tools (PyTorch, Transformers, Librosa) and standard development processes (version control, modular design, experiment tracking, reproducibility) were followed throughout. |
| CO6 | Our capstone project has no harmful societal or environmental impact. The system is not limited to particular demographics—it works for diverse speakers, ages, and genders (though IEMOCAP has balanced male/female representation). This is software-based, requiring no harmful materials. The system uses a microphone for audio input and processes text transcriptions—standard computing devices. This enables faster, more accurate emotion detection than human | **i) Social and Environmental Impact of Engineering [K7]**<br>**K7: Comprehension of engineering in society**: The project shows no harmful activities economically, socially, ecologically, or culturally. Applications in mental health monitoring, conversational AI, and educational technology provide significant societal benefits. Environmental impact is minimal (software-only, standard computing). | **(i) Social and Environmental Impact of Engineering [EP2, EP5, EP6]**<br>**EP2: Range of conflicting requirements**: Tools were selected considering accessibility (open-source libraries), efficiency (GPU-accelerated computation), and reproducibility (standard frameworks).<br>**EP5: Extent of applicable codes**: Standard project management (Gantt charts, task distribution), development practices (version control, code review), and evaluation protocols (LOSO cross-validation, standard metrics) were followed. |

| | | | |
|---|---|---|---|
| | observation alone, supporting mental health monitoring, empathetic conversational AI, and improved human-computer interaction. The system is contactless and privacy-preserving (no video recording, de-identified data). No cultural or legal issues arise—we use publicly licensed data with proper consent. People from all backgrounds can benefit from emotionally intelligent AI systems. | | **EP6: Extent of stakeholder involvement and conflicting requirements**: Stakeholder needs (researchers: reproducibility, developers: efficiency, users: interpretability, ethicists: privacy) were considered in design decisions regarding evaluation protocols, architectural complexity, and transparency mechanisms. |

**CSE400-C**
**Program Outcomes**
**PO7 (Environment and Sustainability)**
Our project is fully software-based and has no physical or environmental impact. Energy use during model training was minimized through mixed-precision training, early stopping, and efficient architectures. The system runs on standard computing resources, and inference requires very little energy, making the solution sustainable and scalable.

**PO8 (Ethics)**: We used the IEMOCAP dataset, which was collected with informed consent, proper compensation, and de-identified data. No new data was collected, avoiding privacy concerns. The system analyzes only user-provided audio and operates transparently. We acknowledge risks of misuse (e.g., surveillance) and emphasize that emotion recognition should always be applied with consent and ethical responsibility.
**PO9 (Individual Work and Teamwork)**: The project relied on collaboration while maintaining individual responsibilities. Team members contributed based on strengths—audio processing, NLP/BERT, fusion models, and evaluation. Regular communication, shared learning, and collective problem-solving helped coordinate the multimodal system effectively and ensured steady progress.
**PO10 (Communication)**: We communicated through meetings, Discord, GitHub, and shared documents. Challenges and results were openly discussed using visualizations like confusion matrices and training curves. Frequent supervisor presentations strengthened clarity, feedback integration, and overall project direction.

**PO11 (Project Management and Finance)**: We applied project management techniques such as WBS, Gantt charts, and RACI assignments. Clear milestones helped maintain progress across dataset prep, baseline models, fusion, and evaluation. Financial needs were minimal due to free academic GPUs, open-source tools, and licensed datasets. Future commercial expansion may require cloud compute and storage costs.
**PO12 (Life-Long Learning)**: This project required learning new concepts such as BERT fine-tuning, multimodal fusion, cross-modal attention, LOSO evaluation, and class imbalance handling. We used papers, tutorials, and framework documentation to build real understanding. The experience strengthened our research ability, experimentation skills, and confidence to tackle advanced AI problems in the future.

| CO | Details | Knowledge Profile (K) | Engineering Problem (EP) |
|---|---|---|---|
| CO7 | Not harming the environment is a priority. Our capstone project has minimal environmental impact. This is software-based, requiring no physical materials or harmful substances. The computing resources used (GPUs for training, CPUs for inference) are standard research infrastructure with manageable energy consumption. We implemented efficiency optimizations: mixed-precision training (reducing computation), early stopping (preventing unnecessary training), efficient architectures (lightweight fusion layers), and model checkpointing (avoiding redundant retraining). The system doesn't require extreme electricity or cooling—standard data center infrastructure suffices. Deployment requires only typical computing devices (servers, edge devices) that users already possess. The system is sustainable and scalable, with minimal incremental environmental cost. | **(i) Societal and environmental contexts [K7]**<br>**K7: Comprehension of engineering in society**: In designing and implementing the project, we considered ethical issues (data privacy, informed consent, transparent operation), public safety (mental health applications must be accurate and reliable), and environmental impact (energy-efficient implementations). We considered economic benefits (improved customer service, mental health support), social impact (accessible empathetic AI), and cultural sensitivity (diverse emotional expressions across cultures). The project was built with minimal financial cost and environmental footprint. | **(i) Societal and environmental contexts [EP2, EP5, EP6]**<br>**EP2: Range of conflicting requirements**: Different fusion strategies have different computational costs—early fusion requires processing raw features (high cost), late fusion processes probability vectors (low cost). We evaluated this trade-off explicitly.<br>**EP5: Extent of applicable codes**: We implemented efficiency optimizations following best practices: mixed-precision training, gradient checkpointing, efficient data loading, early stopping, and lightweight architectures. Standard development processes (version control, code review, documentation) were followed.<br>**EP6: Extent of stakeholder involvement and conflicting requirements**: Stakeholders have varying priorities—researchers need reproducibility (standardized evaluation), developers need efficiency (deployment constraints), users need interpretability (trust in predictions), and ethicists need transparency (understanding system decisions). We addressed these through comprehensive evaluation, efficiency analysis, and interpretable mechanisms (gate weights, attention visualization). |
| CO8 | In planning, we considered ethics carefully. Our application processes emotional speech requiring audio and text data. We | **(i) Ethical principle and practices [K7]**<br>**K7: Comprehension of engineering in** | |

| | | | |
|---|---|---|---|
| | used IEMOCAP, collected with informed consent from professional actors who agreed to research use of their emotional expressions. All participants were compensated and data de-identified. We obtained proper academic licensing through our institution. We collected no data ourselves, avoiding privacy concerns. The dataset is used exclusively for academic research and not shared beyond proper channels. Our system analyzes uploaded audio—it doesn't record secretly or violate privacy. The system's purpose is transparent and clearly communicated. We considered potential misuse (surveillance without consent, emotional manipulation) and emphasize that deployment must involve informed consent, transparent operation, and human wellbeing as the primary goal. Emotion recognition should support mental health and improve interaction, not exploit vulnerabilities or invade privacy. | **society**: We considered ethical principles throughout: informed consent (using properly licensed data), privacy protection (de-identified data, no video recording), transparency (interpretable mechanisms), beneficence (supporting mental health, improving AI), and non-maleficence (avoiding surveillance, manipulation). We documented ethical considerations and emphasized responsible deployment practices. | |
| CO9 | Developing this project was teamwork with individual contributions. Implementation required diverse skills: audio signal processing, deep learning programming, NLP expertise, experimental design, and evaluation. We divided responsibilities based on strengths—some focused on audio processing (feature extraction, LSTM architecture), others on text processing (BERT fine-tuning, preprocessing), and others on fusion (architecture design, training pipelines).<br>Communication was essential for coordinating multimodal systems. Members shared knowledge about unfamiliar areas. Regular meetings ensured alignment. We practiced collaborative problem-solving | | |

| | | | |
|---|---|---|---|
| | when facing challenges (underperforming architectures, class imbalance issues, fusion complexity). We learned that listening to diverse perspectives improves solutions and that clear communication prevents misunderstandings in complex projects. | | |
| CO10 | We have written a comprehensive report on the capstone project. In Chapters 3 (Methodology) and 4 (Design and Implementation), we elaborated on: dataset description and preprocessing, audio feature extraction (mel-spectrograms, MFCCs, normalization), LSTM architecture with attention, BERT fine-tuning strategy, seven fusion architectures (early, late, hybrid, adaptive gating, cross-modal attention, performance-based weighting, learnable weights), training strategies (class weighting, optimization, learning rate scheduling, early stopping), and evaluation protocols (LOSO cross-validation with session separation). In Chapter 5 (Results), we presented comprehensive analysis including per-class performance, confusion matrices, cross-fold variability, fusion effectiveness, gate weight patterns, and comparative evaluation across all approaches. The report provides sufficient detail for reproducibility and clear understanding of our methodology and findings. | | **EA1: Range of resources**: To fulfill our thesis, we involved diverse resources: IEMOCAP dataset (licensed audio and text data), computing infrastructure (GPUs for training), deep learning frameworks (PyTorch, Transformers), audio processing libraries (Librosa), evaluation tools (Scikit-learn), and visualization libraries (Matplotlib).<br>**EA2: Level of interaction**: The main challenge was aligning audio and text data by utterance IDs across 10 cross-validation folds while maintaining session separation. We carefully matched samples across modalities, ensuring proper synchronization for fusion.<br>**EA3: Innovation**: Modular design divided the system into three components: audio module (LSTM with attention for acoustic features), text module (fine-tuned BERT for linguistic features), and fusion module (seven strategies for comparison). This modularity enables future researchers to improve individual components independently or explore new fusion approaches. Our comprehensive comparison of seven fusion strategies in identical experimental conditions is novel, providing empirical evidence about |

| | | | |
|---|---|---|---|
| | | | relative effectiveness.<br>**EA4: Consequences to society and the environment**: Significant positive societal consequences—advancing emotion recognition supports mental health monitoring, improves conversational AI, enhances educational technology, and enables more natural human-computer interaction. Minimal environmental impact (software-only, standard computing).<br>**EA5: Familiarity**: We addressed emotion recognition, which spans psychology (emotion theory), linguistics (emotional language), signal processing (acoustic prosody), and computer science (deep learning, multimodal fusion). This interdisciplinary problem required understanding concepts beyond traditional computer science boundaries. |
| CO11 | This substantial project required hard work and careful management. We applied project management concepts: Work Breakdown Structure (dividing into literature review, data preparation, unimodal baselines, fusion implementations, evaluation, reporting), Gantt charts (timeline planning with milestones), and RACI matrices (responsibility assignment). We set realistic deadlines and milestones, tracking progress systematically. Time management ensured steady progress without last-minute rushes. Financial requirements were minimal—we used free academic computing resources (university GPU cluster), free cloud platforms (Google Colaboratory), open-source software (PyTorch, Librosa, | | |

| | | | |
|---|---|---|---|
| | Transformers), and publicly available datasets (IEMOCAP through academic licensing). No hardware purchases were necessary. For future commercial deployment or large-scale experiments, financial support would be needed for cloud computing credits, high-performance GPUs, storage infrastructure, and potentially commercial dataset licensing. However, for academic research and proof-of-concept, existing resources were sufficient. | | |
| CO12 | Concepts we learned and applied were largely self-initiated. While we had foundational knowledge from coursework, many specific techniques were new: LSTM attention mechanisms, BERT fine-tuning, multimodal fusion architectures, cross-modal attention, adaptive gating, LOSO cross-validation with session separation, handling class imbalance, and interpreting fusion behaviors. We explored academic papers, online tutorials, framework documentation, and implemented concepts ourselves. This process developed research skills—formulating hypotheses, designing experiments, analyzing results, identifying failures, iterating on solutions. We learned to debug complex deep learning systems, interpret unexpected behaviors (like uniform attention weights), and recognize when simpler solutions outperform complex architectures. We gained confidence in tackling novel research problems through systematic experimentation. The skills acquired—deep learning implementation, multimodal system design, experimental methodology, scientific communication—are valuable for future research and industry | | |

| | | | |
|---|---|---|---|
| | careers in AI/ML. This project taught us not just technical skills but also how to approach open-ended research problems, manage uncertainty, and persist through challenges. | | |